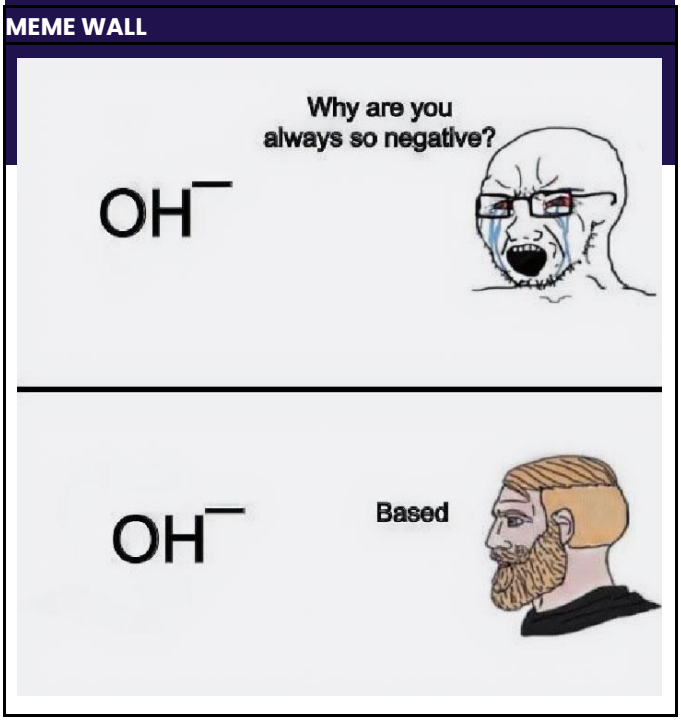
CHEMISTRY LEC | UNITS 5

MEME WALL



REDOX REACTIONS

Introduction

- **Oxidation-reaction (redox) chemical reaction**: Chemical reaction in which there is a **transfer of electrons** from one reactant to another reactant.
- **Non-oxidation-reduction (non-redox) chemical reaction**: Chemical reaction in which there is **no NET electron transfer** from one reactant to another reactant.

Oxidation-Redution Reactions

- Often called **"redox" reactions**
- Electrons are transferred between the reactants
  - **OXIDATION/REDUCING AGENT:** One substance in oxidized, loses electrons
  - **REDUCTION/OXIDIZING AGENT:** Another substance is reduced, gains electrons
- Oxidation numbers change during the reaction.

REMEMBER OIL RIG (FOR REDOX)

- **O**xidation **I**s **L**oss (of electrons)
- **R**eduction **I**s **G**ain (of electrons)

Oxidation Number (Oxidation State)

- A number that represents the charge that an atom appears to have when the electrons in each bond is participating in are assigned to the more electronegative of the two atoms involved in the bond.

Rules for Determining Oxidation Numbers

1. **Elemental State** = 0 (Mg, Na, O2 = 0)
2. **Monatomic Ion** = charge of the ion (Mg^{2+} + 2 | Cl^{-} = -1)
3. **Group 1A** = +1 | **Group IIA** = +2
4. **Hydrogen** = +1
5. **Oxygen** = -2 but - 1 (for hydrogen peroxide & peroxide ion)
6. **Binary Molecular Compounds**: The more electronegative element, is assigned a negative oxidation number equal to its charge in binary ionic compounds.
7. **For compounds,** sum of oxidation numbers = 0;
8. **Polyatomic ions,** sum of oxidation numbers = charge of the ion.

Balancing Redox Reactions

- **Electrons (charge)** must be balanced as well as number and types of atoms.

Consider this net ionic reaction:

$Al(s) + Ni^{2+}(aq) \rightarrow Al^{3+}(aq) + Ni(s)$

– The reaction appears balanced as far as number and type of atoms are concerned, but look closely at the charge on each side.

Step 1: Divide reaction into two half-reactions

From this: $Al(s) + Ni^{2+}(aq) \rightarrow Al^{3+}(aq) + Ni(s)$

Step 1.1
Oxidation $Al(s) \rightarrow Al^{3+}(aq)$
Reduction $Ni^{2+}(aq) \rightarrow Ni(s)$

- Add charges (electrons) on a side depending on where it is needed to make the charges at both sides equal.

To: Step 1.2
Oxidation $Al(s) \rightarrow Al^{3+}(aq) + 3e^-$
Reduction $Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$

- As seen above, Al's charge from the left side is 0, while Al's charge from the right side is 3+. We must put a 3e- at the right side to make it equal to the left side. While we add 2e- to the left side of Ni^{2+}

Step 2: Multiply by a common factor to equalize electrons (the number of electrons lost must equal number of electrons gained)

- We find a common factor so we can cancel the charges since they are not equal.

2 [Al(s) → Al3+(aq) + 3e-]
3 [Ni2+(aq) + 2e- → Ni(s)]

Step 3: Cancel electrons and write balanced net ionic reaction

- **Cancel out 6e-**

2Al(s) → 2Al3+(aq) + 6e-
3Ni2+(aq) + 6e- → 3Ni(s)

- You are left with:

2Al(s) + 3Ni2+(aq) → 2Al3+(aq) + 3Ni(s)

Which is now a Balanced Redox Reaction.
Yehey!!

Example #2:

Zn(s) + CuSO4(aq) -> ZnSO4(aq) + Cu(s)

Zn(s) + Cu2+(aq) -> Zn2+(aq) + Cu(s)

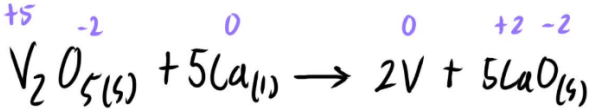
- Zinc is losing 2 electrons and is oxidized.
  - *Reducing agent*
  - Zn(s) -> Zn2+(aq) + 2e-
- Copper ions are gaining the 2 electrons.
  - Oxidizing agent
  - Cu2+(aq) + 2e- -> Cu(s)

TYPES OF REDOX REACTIONS

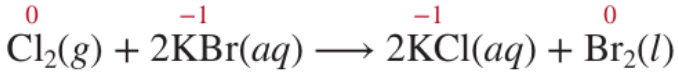
1. Displacement Reactions

- A common reaction where an **active metal** replaces (DISPLACES) a **metal ion from a solution** (same for halogens).
- The **Activity Series of Metals or Halogens** is useful in order to predict the outcome of the reaction.
- All **single-replacement** reactions are redox reactions, but **double-replacement** are generally not.

Metal Displacement



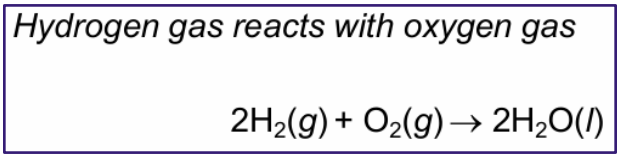
Halogen Displacement

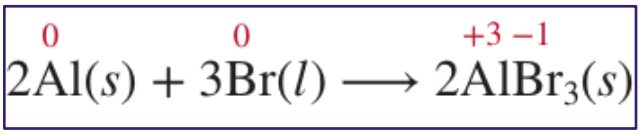


2. Combination Reactions

- A combination reaction is a reaction in which **two** or **more** substances **combine** to form a **product**.
- Not all combination reactions are redox reactions.

Hydrogen gas reacts with oxygen gas

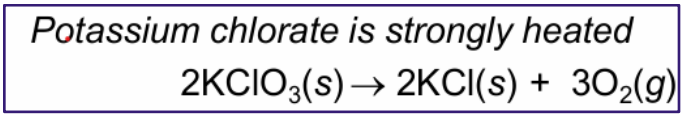


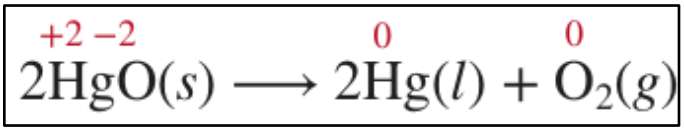


3. Decomposition Reactions

- Decomposition reaction is the **breakdown** of a compound into **two or more components**.
- Not all decomposition reactions are redox reactions.

Potassium chlorate is strongly heated





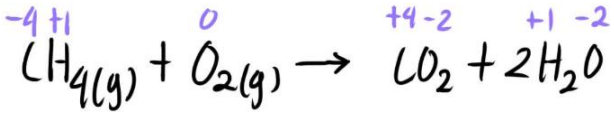
4. Disproportionation Reactions

- An element in one oxidation state is **simultaneously oxidized** and **reduced**.
- One reactant in a disproportionation reaction always contains an **element** that can have at **least three oxidation states**.
- The element itself is in an **intermediate oxidation state**, that is, both **higher** and **lower** oxidation states exist for that element in the **products**.

$$\overset{-1}{2H_2O_2}(aq) \longrightarrow \overset{-2}{2H_2O}(l) + \overset{0}{O_2}(g)$$

5. Combustion Reactions

- A reaction in which a substance reacts with **oxygen**, usually with the release of **heat and light** to produce a **flame**.
- All combustion reactions are redox processes.



HYDROLYSIS OF SALTS (ACID-BASE PROPERTIES OF SALTS)

- A salt is an **ionic compound** formed by the **reaction between an acid and a base.**
  - Salts are **strong electrolytes** that **completely dissociate** into ions in water.
- **Salt hydrolysis** describes the reaction of an anion or a cation of a salt, or both, with water. It usually affects the pH of a solution.
- The word **"hydrolysis"** is derived from the Greek words ***hydro***, meaning "water," and ***lysis***, meaning "to split apart."

SALTS THAT PRODUCE NEUTRAL SOLUTIONS

- Salt contains an **alkali metal or alkaline earth** metal ion (except Be2+) and the conjugate base of a strong acid (e.g Cl-, Br-, and NO3-).
- For instance, when $NaNO_3$, a salt formed by the reaction of NaOH with HNO_3, dissolves in water, it dissociates completely as follows:

$$NaNO_3(s) \xrightarrow{H_2O} Na^+(aq) + NO_3^-(aq)$$

The hydrated Na+ ion neither donates nor accepts H+ ions. The NO3- ion is the conjugate base of the strong acid HNO3, and it has no affinity for H+ ions. Consequently, a solution containing Na + and NO3- ions is neutral, with a pH of about 7.

SALTS THAT PRODUCE BASIC SOLUTIONS

- The solution of a salt derived from a **strong base** and a **weak acid** is **basic**.
- For example, the dissociation of sodium acetate (CH3COONa) in water is given by:

$$NaCH_3COOH\ (s) \xrightarrow{H_2O} Na^+\ (aq) + CH_3COO^-\ (aq)$$

$$CH_3COO^-\ (aq) + H_2O\ (l) \rightleftharpoons CH_3COOH\ (aq) + OH^-\ (aq)$$

- Because this reaction **produces OH- ions**, the sodium acetate solution will be **basic**.

SALTS THAT PRODUCE ACIDIC SOLUTIONS

- Salts derived from a **strong acid** and a **weak base.**
- When a salt derived from a **strong acid** such as **HCl** and a **weak base** such as **NH3** dissolves in water, the solution becomes **acidic**. For example, consider the process:

$$NH_4Cl\ (s) \xrightarrow{H_2O} NH_4^+\ (aq) + Cl^-\ (aq)$$

$$NH_4^+\ (aq) \rightleftharpoons NH_3\ (aq) + H^+\ (aq)$$

*Note that this reaction also represents the **hydrolysis** of the NH4+ ion. Because **H+ ions are produced**, the **pH** of the solution **decreases**.*

TYPE OF SOLUTION	CATIONS	ANIONS	pH
ACIDIC	From weak bases NH_4^+, Al^{3+}, Fe^{3+}	From strong acids Cl^-, Br^-, I^-, NO_3^-, ClO_4^-	<7
BASIC	From strong bases Group 1 and Group 2, but not Be^{2+}	From weak acids F^-, NO_2^-, CN^-, CH_3COO^-	>7
NEUTRAL	From strong bases Group 1 and Group 2, but not Be^{2+}	From strong acids Cl^-, Br^-, I^-, NO_3^-, ClO_4^-	=7

CONCENTRATION OF SOLUTIONS

Concentration

- The **amount of solute** present in a specified amount of **solution**

Methods of expressing concentration

1. Percent concentration: Percent by Mass (mass-mass%), Percent by volume (volume-volume %), and Mass-volume percent
2. Molarity concentration
3. Molality concentration
4. Normality concentration

COMPARISON OF CONCENTRATED AND DILUTE SOLUTION

- They are qualitative expressions of concentration
- **Concentrated:** higher ratio of solute to solvent
- **Dilute:** smaller ratio of solute to solvent

PERCENT CONCENTRATION

1. PERCENT BY MASS

- Often used to express the concentration of **commercial aqueous reagents.**
- IUPAC terminology: it is the mass concentration.
- **UNITS: % w/w = grams per 100 g**

$$\textbf{Percent by mass} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100\ \%$$

$$\text{Mass of solution} = \text{mass of solute} + \text{mass of solvent}$$

Example:

1. What is the percent by mass concentration of glucose in a solution made by dissolving 5.5g of glucose in 78.2g of water?

$$\%\ \textit{mass of glucose}\ \ \frac{\text{mass of solute}}{\text{mass of solution}} = \frac{5.5\text{g}}{5.5\text{g} + 78.2\text{g}} \times 100 = 6.571087216\ \%^{\text{w}}/_{\text{w}} \Rightarrow \boxed{6.6\ \%^{\text{w}}/_{\text{w}}}$$

2. PERCENT BY VOLUME

- Commonly used to specify the concentration of a solution prepared by diluting a pure liquid compound with another liquid.
- IUPA terminology: volume concentration
- Both solute and solvents are **liquids.**
- **UNITS: % v/v = mL per 100 mL**

$$\textbf{Percent by volume} = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100\ \%$$

Example:

1. Calculate the volume percent of solute in 30.0 mL of methyl alcohol in enough water to give 435 mL of solution.

$$\%\ \textit{volume of solute}\ \ \frac{\text{volume of solute}}{\text{volume of solution}} = \frac{30.0\ \text{mL}}{435\ \text{mL}} \times 100 = 6.896551724\ \%^{\text{v}}/_{\text{v}} \Rightarrow \boxed{6.90\ \%^{\text{v}}/_{\text{v}}}$$

3. MASS-VOLUME PERCENT

- Often used to indicate the composition of dilute aqueous solutions of solid reagents.
- **UNITS: % w/v = grams per 100 mL**
- Units should **always be defined** since both are in different dimensions.

Mass- … **0%**

Exampl

1. … tion … are

the problem states that we have 5.00 % glucose in the solution. this means for every 100mL of solution, we have 5.00g of glucose.

$$\%\ ^{\text{m}}/_{\text{v}}\ \ \frac{5.00\ \text{g}}{100\ \text{mL}} = \frac{\text{x g}}{250\ \text{mL}} \quad \text{cross multiply}$$

$$100\text{x} = (5.00\ \text{g})(250\ \text{mL}) \quad \text{solve for x}$$

$$\frac{100\text{x}}{100} = \frac{(5.00\ \text{g})\ (250\ \cancel{\text{mL}})}{100\ \cancel{\text{mL}}}$$

$$\text{x} = \boxed{12.5\ \text{g glucose}}$$

2. Describe the preparation of 500 mL of 5.25% (V/V) aqueous ethanol (C2H5OH, 46.1 g/mol)

500 mL of 5.25% (v/v) aqueous ethanol.

$$5.25\%\ (\text{v/v})\ \text{C}_2\text{H}_5\text{OH} = \frac{5.25\ \text{mL C}_2\text{H}_5\text{OH}}{100\ \text{mL soln}}$$

$$\frac{5.25\ \text{mL C}_2\text{H}_5\text{OH}}{100\ \text{mL soln}} \times 500\ \text{mL soln} = 26.3\ \text{mL C}_2\text{H}_5\text{OH}$$

Dilute 26.3 mL ethanol with enough water to give a final volume of 500 mL.

Table 8.3 - Conversion Factors Obtained From Percent Concentration Units

Percent Concentration	Meaning in Words	Conversion Factors		
15%(m/m) NaCl solution	There are 15 g of NaCl in 100 g of solution.	$\frac{15\ \text{g NaCl}}{100\ \text{g solution}}$	and	$\frac{100\ \text{g solution}}{15\ \text{g NaCl}}$
6%(v/v) methanol solution	There are 6 mL of methanol in 100 mL of solution.	$\frac{6\ \text{mL methanol}}{100\ \text{mL solution}}$	and	$\frac{100\ \text{mL solution}}{6\ \text{mL methanol}}$
25%(m/v) sucrose solution	There are 25 g of sucrose in 100 mL of solution.	$\frac{25\ \text{g sucrose}}{100\ \text{mL solution}}$	and	$\frac{100\ \text{mL solution}}{25\ \text{g sucrose}}$

Sample Exercise: try niyo solve guys

1. A solution having a mass percent concentration of 10% contains? **10g of solute and 90g of solvent**

2. How many grams of sodium chloride are needed to prepare 150 mL of 0.90% (m/v) sodium chloride? **1.4g**

MOLARITY CONCENTRATION (M)

- The number of moles of that species that is contained in **1 liter of the solution** (not 1L of the solvent).
- The number of millimoles of solute per milliliter of solution.
- mol/L = mmol/mL
- **UNITS: M (molar) with dimensions mol/L or mol L^-1**

$$\text{Molarity (M)} = \frac{\text{moles of solute}}{\text{liters of solution}}$$

Examples:

1. Determine the molarities of 4.35 moles of KMnO4 dissolved in enough water to give 750 mL of solution.

Solution

a. The number of moles of solute is given in the problem statement.

$$\text{Moles of solute } (KMnO_4) = 4.35 \text{ moles}$$

The volume of the solution is also given in the problem statement, but not in the right units. Molarity requires liters for the volume units, and milliliters of solution are given. Making the unit change yields

$$750 \text{ mL} \times \left(\frac{10^{-3} \text{ L}}{1 \text{ mL}}\right) = 0.750 \text{ L}$$

The molarity of the solution is obtained by substituting the known quantities into the equation

$$M = \frac{\text{moles of solute}}{\text{liters of solution}}$$

which gives

$$M = \frac{4.35 \text{ moles } KMnO_4}{0.750 \text{ L solution}} = 5.80 \frac{\text{moles } KMnO_4}{\text{L solution}}$$

Note that the units for molarity are always moles per liter.

2. Calculate the molarity of a solution made by dissolving 23.4 g of sodium sulfate (Na2SO4) in enough water to form 125mL of solution. (Given MM Na2SO4 = 142.00 g/mol).

Solve The number of moles of Na_2SO_4 is obtained by using its molar mass:

$$\text{Moles } Na_2SO_4 = (23.4 \text{ g } Na_2SO_4)\left(\frac{1 \text{ mol } Na_2SO_4}{142 \text{ g } Na_2SO_4}\right) = 0.165 \text{ mol } Na_2SO_4$$

Converting the volume of the solution to liters:

$$\text{Liters soln} = (125 \text{ mL})\left(\frac{1 \text{ L}}{1000 \text{ mL}}\right) = 0.125 \text{ L}$$

Thus, the molarity is

$$\text{Molarity} = \frac{0.165 \text{ mol } Na_2SO_4}{0.125 \text{ L soln}} = 1.32 \frac{\text{mol } Na_2SO_4}{\text{L soln}} = 1.32 \, M$$

Note: Conversions, 1L = 1000 mL * 1mol = 1000mmol

CONVERT PERCENT CONCENTRATION USING SPECIFIC GRAVITY

Example: Calculate the molar concentrations of HNO3 (63.00 g/mol) in a solution that has a specific gravity of 1.42 and 70.5% of HNO3 (w/w).

Let us first calculate the mass of acid per liter of concentrated solution

$$\frac{\text{g HNO}_3}{\text{L reagent}} = \frac{1.42 \text{ kg reagent}}{\text{L reagent}} \times \frac{10^3 \text{ g reagent}}{\text{kg reagent}} \times \frac{70.5 \text{ g HNO}_3}{100 \text{ g reagent}} = \frac{1001 \text{ g HNO}_3}{\text{L reagent}}$$

Then,

$$c_{HNO_3} = \frac{1001 \text{ g HNO}_3}{\text{L reagent}} \times \frac{1 \text{ mol HNO}_3}{63.0 \text{ g HNO}_3} = \frac{15.9 \text{ mol HNO}_3}{\text{L reagent}} \approx 16 \text{ M}$$

Example: Consider separate solutions of NaOH and KCl made by dissolving 16 g of each solute in 250 mL of solution. Calculate the concentration of each solution. (Given mm: NaOH = 40.00g; KCl = 74.55 g/mol)

Find the mols of NaOH and KCl before calculating the molarity (*convert 250mL into liters*)

$$\triangleright \text{ mol NaOH} = \frac{16\text{g NaOH}}{40.00\text{g/mol NaOH}} = 0.4 \text{ mol NaOH}$$

$$\triangleright \text{ M} = \frac{0.4 \text{ mol NaOH}}{0.250\text{L solution}} = \boxed{1.60\text{M NaOH}}$$

$$\triangleright \text{ mol KCl} = \frac{16\text{g KCl}}{75.44\text{g/mol KCl}} = 0.2146 \text{ mol KCl}$$

$$\triangleright \text{ M} = \frac{0.2146 \text{ mol KCl}}{0.250\text{L solution}} = \boxed{0.858\text{M KCl}}$$

MOLALITY CONCENTRATION (m)

- The number of moles of solute in 1000 g or 1 kg of solvent rather than the final solution.
- **UNITS: m (molality) with dimensions mol/kg**

$$\text{Molality} = \frac{\text{Weight of the solute}}{\text{Molecular weight x kilogram of solvent}}$$

Unit: moles/kilogram

Example:

1. What is the molality of a solution if 127 g of NaCl is dissolved in 1000.0 g of distilled water? (Given MM = 58.44 g/mol)

$$m = \frac{\text{weight of solute}}{\text{mw x Kg of solvent}}$$

$$m = \frac{127 \text{ g NaCl}}{(58.44 \frac{\text{g NaCl}}{\text{mol NaCl}})(1 \text{ Kg solvent})}$$

m = 2.17 mol/ kg

2. A solution is made by dissolving 4.25 g glucose into 250.00 g of water. Calculate the molality of glucose in the solution. (MM = 180.18 g/mol glucose) **ANSWER**: 0.0994m of C6H12O6 (Glucose)
3. Convert a 0.850 molal solution of NaCl to percent (w/w). (MM NaCl = 58.44 g/mol)

NORMALITY CONCENTRATION (N)

- Unit of concentration of a chemical solution expressed as gram equivalent weight of solute per liter of solution.
- A defined equivalence factor must be used to express concentration.
- **UNITS: N (normality), eq/L, or med/mL**
- **Equivalent**: the quantity of acid in an acid base reaction 1 mol of H+ ions or the quantity of base that reacts with 1 mol H+ ions.

$$\text{Normality} = \frac{\text{equivalent weight of solute}}{\text{liters of solution}}$$

1 equivalent of acid reacts with 1 equivalent of base.

EQUIVALENT WEIGHT (EW)

- **Equivalent** – the amount of a substance that reacts with or replaces **1 mole of H^+**.
- ▪ Depends on the **charge/valence** of the substance.
  ▪ **Acid:** amount that provides **1 mole of replaceable H^+**.
  ▪ **Base or salt:** amount that reacts with **1 mole of H^+**.

$$EW = \frac{MW}{\text{Valence}}$$

Where:

• **MW** = molar mass (g/mol)

• **Valence/n-factor** = number of H^+ involved in the reaction, or the relevant total charge

QUESTION | HOW TO FIND THE VALENCE?

Acid: count the replaceable H^+.
Base: count the H^+ it can react with.
Salt/ion: use the relevant total charge.

	valence	EW
NaCl	1	58.44
$BaCl_2$	2	104.12
K_2CO_3	2	69
$CaSO_4$	2	68
$AlCl_3$	3	44
$Fe_2(SO_4)_3$	6	66.6

Example:

1. What is the molarity for a solution with 0.587g of NaOH (MM = 40.00 g/mol) in 25.7 mL?

$$N = \frac{g / (\text{molar mass} / \text{Eq. wt.})}{L} \qquad N = \frac{\text{weight of solute}}{\text{EW x L of solution}}$$

$$N = \frac{(0.587\ \cancel{g} / 40.00\ \cancel{g}/\cancel{mol} / 1\text{eq. wt.}/\cancel{mol})}{0.0257\ L}$$

$$N = 0.571\ Eq/L$$

SAMPLE PROBLEMS

Problem: Find the equivalent weight of H_2SO_4. MW = 98.08 g/mol.
Step 1: H_2SO_4 has 2 replaceable H^+ → valence = 2.
Step 2: Substitute into the formula:

$$EW = \frac{98.08\ g/mol}{2}$$

EW = 49.04 g/eq

Problem: A compound has a molar mass of 138.20 g/mol and a valence of 2. Determine its equivalent weight.
Step 1: MW = 138.20 g/mol; valence = 2.
Step 2:

$$EW = \frac{138.20\ g/mol}{2}$$

EW = 69.10 g/eq

Problem: What is the normality of a solution containing 0.321 g Na_2CO_3 in 250.0 mL of solution? MW = 105.99 g/mol.

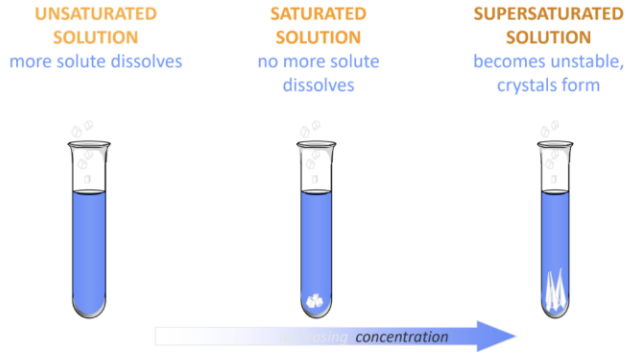
Step 1 — Find the valence / n-factor
CO_3^{2-} has a charge of 2− → n-factor = 2.
Step 2 — Calculate equivalent weight
EW = MW ÷ valence
EW = 105.99 ÷ 2 = 52.995 g/eq
Step 3 — Convert mL to L
250.0 mL ÷ 1000 = 0.2500 L
Step 4 — Calculate normality
N = g ÷ (EW × L)
N = 0.321 g ÷ [(52.995 g/eq)(0.2500 L)]
N = 0.02423 eq/L
Answer: N = 0.0242 N

Problem: The molar mass of K_2CO_3 is 138.20 g/mol. What is its equivalent weight?

Step 1 — Find the valence / n-factor
CO_3^{2-} has a charge of 2− → valence = 2.
Step 2 — Apply the EW formula
EW = MW ÷ valence
EW = 138.20 ÷ 2
EW = 69.10 g/eq

SOLUBILITY

- The **maximum proportion of the solute that can dissolve** in a given amount of the solvent, usually expressed in grams of solute in 100 grams of solvent,
- Refers to the **solubility of the solute** in the solvent.
- Based on a saturated solution:



FACTORS AFFECTING SOLUBILITY

- **Temperature -** for around 95% of solid solutes, solubility increases with temperature. However in gaseous solutes, they become less soluble in water and more soluble in organic solvents.
- **Pressure -** the solubility of a gas is directly proportional to the pressure of the gas (Henry's Law)
- **Polarity -** "like dissolves like" a solute will dissolve better in a solvent with a similar polarity

----------END OF CHAPTER----------

+IMPORTANT FORMULAS

$$\textit{Percent by mass} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100\,\%$$

$$\text{Mass of solution} = \text{mass of solute} + \text{mass of solvent}$$

$$\textit{Percent by volume} = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100\,\%$$

$$\textit{Mass-volume percent} = \frac{\text{mass of solute(g)}}{\text{volume of solution(mL)}} \times 100\%$$

$$\text{Molarity (M)} = \frac{\text{moles of solute}}{\text{liters of solution}}$$

$$\textit{Molality} = \frac{\textit{Weight of the solute}}{\textit{Molecular weight x kilogram of solvent}}$$

Unit: moles/kilogram

$$\text{Normality} = \frac{\text{equivalent weight of solute}}{\text{liters of solution}}$$

1 equivalent of acid reacts with 1 equivalent of base.

$$EW = \frac{MW}{\text{Valence}}$$

MOCK QUIZ

TEST 1. TRUE OR FALSE (MC)

1. When chlorine forms Cl- ions, its oxidation number increases
2. A redox reaction always involves a change in oxidation numbers.
3. In electrochemical cells, oxidation occurs at the cathode.
4. An increase in oxidation number indicates that electrons were gained.
5. The stronger the attractions between solute and solvent molecules, it becomes harder for a solute to dissolve them in solvent.
6. Most solid solutes become less soluble in water with decreasing pressure.
7. If additional solute is added to a saturated solution, the solution becomes supersaturated.
8. All ionic substances are soluble in water due to attraction of their positive and negative charges to the dipoles of water.
9. Molality (m) is defined as the number of moles of solute contained in exactly 1 liter of the final solution.
10. For neutral compounds, the sum of all the oxidation numbers of its constituent atoms must equal zero.
11. A non-oxidation-reduction reaction is a chemical reaction in which there is a net electron transfer from one reactant to another,
12. In a redox reaction, the substance that is oxidized loses electrons and acts as a reducing agent.
13. Elements in their free, uncombined state always have an oxidation number of zero.
14. The oxidation number for hydrogen is -1 in most hydrogen-containing compounds.
15. The oxidation number of oxygen in oxygen-containing compounds is always -2 under all circumstances.
16. For polyatomic ions, the sum of the oxidation numbers must equal the net charge of the ion.
17. A salt derived from a strong acid and a weak acid produces a basic solution.

18. A salt derived from a strong acid and a weak base produces an acidic solution.
19. Solubility of solid solutes generally increases as temperature rises for about 95% of solid solutes.
20. According to Henry's Law, the solubility of a gas is indirectly proportional to the pressure of that gas.
21. Polar solutes dissolve best in nonpolar solvents following the rule "like dissolves like"
22. In mass-percent concentration (*w/w*), the mass of the solution is calculated by subtracting the mass of the solute from the mass of the solvent.
23. Normality is expressed as the gram equivalent weight of solute per liter of solution.

TEST II. BALANCING THE EQUATION BY SHOWING THEIR HALF0REACTIONS

1. $Al(s) + Ni^{2+}(aq) \rightarrow Al^{3+}(aq) + Ni(s)$
2. $H_2(g) + O_2(g) \rightarrow H_2O(l)$
3. $KClO_3(s) \rightarrow KCl(s) + O_2(g)$

TEST III. MULTIPLE CHOICE/PROBLEM SOLVING

1. What term describes a chemical reaction in which electrons are transferred between reactants?

a. Precipitation reaction
b. Oxidation-reduction (redox) reaction
c. Neutralization reaction
d. Non-redox reaction

2. In the reaction of magnesium burning in oxygen, what happens to magnesium?

a. It gains electrons and is reduced
b. It loses electrons and is oxidized
c. Its oxidation number decreases
d. It acts as an oxidizing agent

3. What is the oxidation number of chromium (Cr) in potassium dichromate ($Kr_2Cr_2O_7$)?

a. +3
b. +6
c. +7
d. -2

4. Which of the following metals is the most reactive according to the metal activity series?

a. Copper (Cu)
b. Zinc (Zn)
c. Lithium (Li)
d. Gold (Au)

MASS PERCENT

5. What is the percent by mass concentration of glucose in a solution made by dissolving 5.5g of glucose in 78.2 of water?

a. 6.9%
b. 7.3%
c. 13.2%
d. 6.6%

6. A commercial aqueous sodium chloride solution has a mass percent concentration of 12.0% (w/w). What is the mass of sodium chloride dissolved in 250.0g of the solution?

A. 30.0g NaCl
B. 52.0g NaCl
C. 31.2g NaCl
D. 23.4g NaCl

VOLUME PERCENT

7. Describe the preparation of 500mL of a 5.25% (v/v) aqueous ethanol solution. How many milliliters of pure ethanol are required?

a. 67.7 mL ethanol
b. 56.32 mL ethanol
c. 26.3 mL ethanol
d. 13.3mL ethanol

8. A technician uses 12.5mL of pure acetone to prepare a 2.50% (v/v) acetone solution. What is the total final volume of the solution in milliliters?

a. 500 mL
b. 250 mL

c. 325 mL
d. 870 mL

MASS-VOLUME PERCENT

9. A nurse prepares a 5.00 (m/v) glucose solution. How many grams of glucose are needed to prepare 250.0 mL of the solution?

a. 13.6g C6H12O6
b. 12.5g C6H12O6
c. 11.2g C6H12O6
d. 23g C6H12O6

GENERAL

10. What is the normality of a 0.0521 M H_3PO_4 solution (Valence = 3).
a. 0.253 N
b. 0.134 N
c. 1.653 N
d. 0.156 N

11. How much mass of aluminum sulfate $Al_2(SO_4)_3$ **(mm = 342.15g/nol)** is needed to prepare 100 mL of a 2.00 N solution with respect to aluminium ions (Al^{3+})?
a. 10.9 g
b. 3.44 g
c. 86.7 g
d. 11.4 g

12. What is the molality of a solution if 127g of NaCl is dissolved in 1000.0 (1kg) of distilled water? (MM = 58.44 g/mol)
a. 3.41 mol/kg
b. 2.17 mol/kg
c. 5.54 mol/kg
d. 2.23 mol/kg

13. Calculate the molarity of a solution made by dissolving 23.4g of sodium sulfate (Na_2SO_4) in enough water to form 125mL of solution (Molar mass Na_2SO_4 = 142.00 g/mol).
a. 1.32 M
b. 0.94 M
c. 2.58 M
d. 1.78 M

14. A solution is made by dissolving 4.25g of glucose ($C_6H_{12}O_6$) into 250.00 g of H2O (MM = 180.18 g/mol). What is the molality of glucose?
a. 0.00564 m
b. 0.0843 m
c. 0.0944 m
d. 0.0478 m

15. How many grams of urea (CON_2H_4, mm = 60.06 g/mol) must be dissolved in 500.0g of H2O to prepare a 0.500 m solution?
a. 16.0 g
b. 15.0 g
c. 14.0 g
d. 17.0 g

ANSWER KEY

TEST 1. TRUE OR FALSE

1. False (it decreases)
2. True
3. False (it occurs at the anode)
4. False (it indicates a *loss* of electrons. Like how gaining electrons = decrease in oxidation no.)
5. False (Hydration enhances solubility/ like dissolves like)
6. False
7. False
8. False
9. False (1 kg instead of 1L)
10. True
11. False
12. True
13. True
14. False (+1 is the correct oxidation no.)
15. False
16. True
17. False
18. True
19. True
20. False
21. False
22. False (Mass of solution = mass of solute + mass of solvent)
23. True

TEST II. BALANCING

1. $2Al(s) + 3Ni^{2+}(aq) \rightarrow 2Al^{3+}(aq) + 3Ni(s)$
2. $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
3. $2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$

TEST II. MULTIPLE CHOICE/PROBLEM SOLVING

1. B (redox reaction)
2. B
3. B
4. C
5. D. 6.6%
6. A. 30.0G NaCl

Formula for Mass of Solute: #6

$$\frac{\text{percent } (w/w) \times \text{mass of solution}}{100\%}$$

7. C. 26.3mL ethanol
   Formula for solute volume: #7

$$\frac{\text{percent } (v/v) \times \text{volume of solution}}{100}$$

8. A. 500 mL
   Formula for solution volume: #8

$$\frac{\text{volume of solute} \times 100}{\text{percent } (v/v)}$$

9. B. 12/5g C6H12O6
10. D. 0.156 N
11. D. 11.4 g
12. B. 2.17 mol/kg
13. A. 1.32 M
14. C. 0.0944 m
15. B. 15.0g

CHEMISTRY LEC I UNITS 6 & 7

MEME WALL



CHEMICAL EQUILIBRIUM I

Chemical equations are typically represented to follow one direction only. Take for example;

$$2H_2\,(g) + O_2\,(g) \rightarrow 2H_2O\,(l)$$

However, chemical reactions are not limited to a one-direction flow. Rather, **all chemical reactions are reversible in theory**.

E.g.

$$2H_2O\,(l) \rightarrow 2H_2\,(g) + O_2\,(g)$$

When reactants turn into products, this is called a **forward reaction**, whereas a **reverse reaction** occurs when products turn into reactants.

In **isolated systems**, we find that not all reactants are converted into products. Instead, some reactants are left behind. This phenomenon is called a **chemical equilibrium**, of which, both reactants and products are created at an equal rate.

A common demonstration of chemical equilibrium is the formation of nitrogen dioxide (NO_2 , a brown gas) from dinitrogen tetraoxide (N_2O_4 , a colorless gas), given by the equation;

$$N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g)$$

In an isolated chamber, the color of the gas becomes darker as N_2O_4 is converted to NO_2.

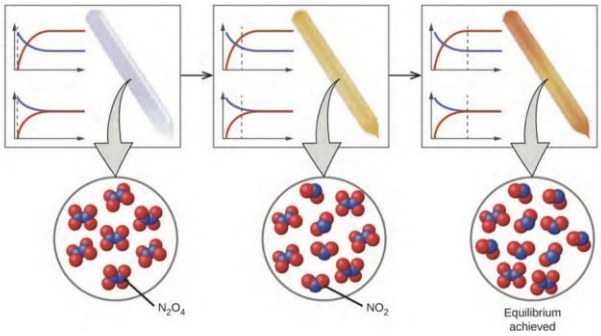


Figure 1. A mixture of NO_2 and N_2O_4 as they shift towards chemical equilibrium

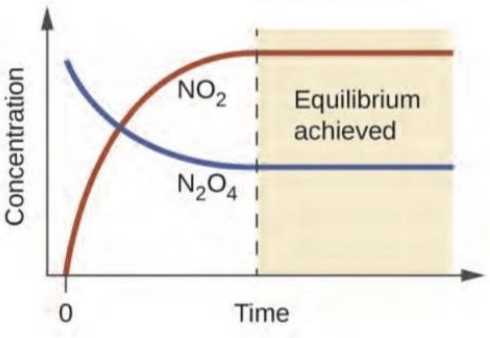


Figure 2. Graph showcasing the change of concentration between NO_2 and N_2O_4 overtime.

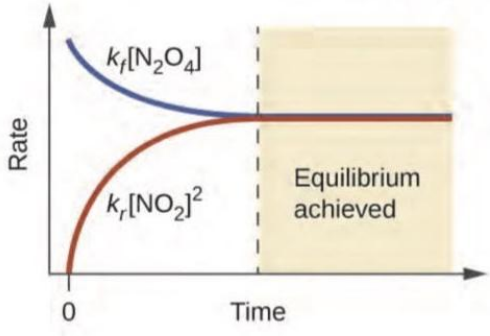


Figure 3. Graph showcasing the rate of reaction of NO_2 and N_2O_4 overtime.

Notice in Figure 3 and 2, that **equilibrium is only achieved once concentration and rate of reaction**

become constant. However, an equal rate of change does not necessarily mean an equal concentration between both reactants and products.

THE EQUILIBRIUM CONSTANT

The same ratio of concentrations is achieved at equilibrium, regardless of the starting composition of reactants and products.

For a general reaction

$$aA + bB \Leftrightarrow cC + dD$$

,

The equilibrium constant is

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

.

- K is constant and depends on stoichiometry, not mechanism.
  - Balancing equations is especially important because the coefficients are used as the exponents.
- Pure solids and liquids are not included if the concentrations are low.
  - Ex. water, a pure solvent, is disregarded.
- **[ ]**-concentration of substance (in M)

Ex.

$$Na_2O_{2(s)} \leftrightarrow Na_{(s)+}O_{2(g)}$$

Since Na_2O_2 and Na are solids, they are not included. Therefore, K in this instance is $K=[O_2]$.

MAGNITUDE OF EQUILIBRIUM CONSTANTS:

K>>1 (far from 1)	Essentially all products (equilibrium lies to the right)
K>1 (close to 1)	product-favored
K=1	reactant=products
K<1 (close to 1)	reactant-favored
K<<1 (far from 1)	Essentially all reactants (equilibrium lies to the right)

LE CHÂTELIER'S PRINCIPLE

We have solidified the fact that reactions proceed in both directions. Mathematically, a reaction is at equilibrium when;

$$Q = K$$

, where Q is the reaction quotient, and K is the equilibrium constant.

When a reaction is already in equilibrium, it will stay in equilibrium. However, **a chemical equilibrium that is disturbed (or, in other words, subjected to stress), will return to equilibrium by counteracting the disturbance**. This is called **Le Châtelier's Principle.** To counteract the disturbance, a chemical reaction may shift towards the products, or shift towards the reactants.

EFFECT OF CHANGE IN CONCENTRATIONS

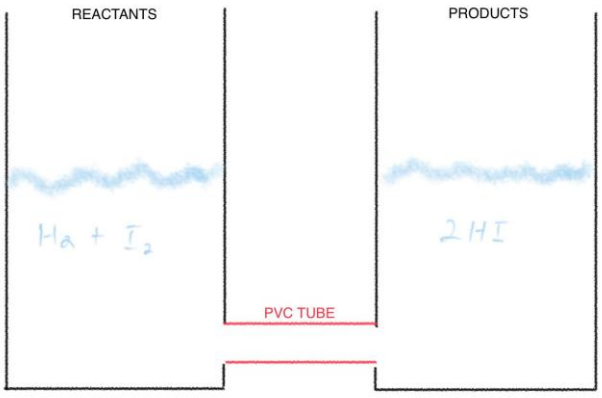
A chemical reaction in equilibrium may be disturbed/stressed by **changing the concentration of reactants/products** by adding/removing one or more reactants/products.

GLASS OF WATER ANALOGY:

Take for instance, the reaction;

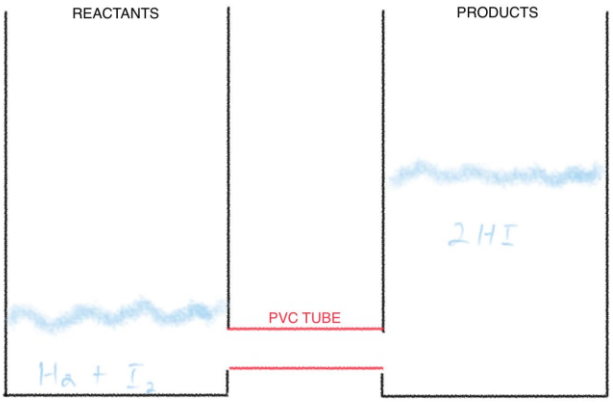
$$H_2\,(g) + I_2\,(g) \rightleftharpoons 2HI$$

wherein, the reaction is currently at equilibrium. Using the glass of water analogy, the reaction would look like this:

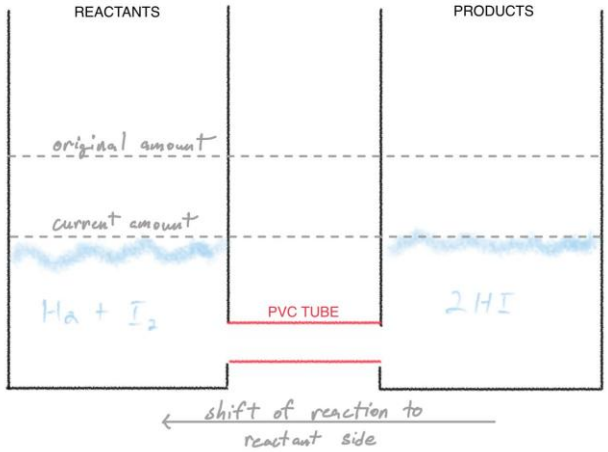


The image shows that the reactants and products are currently at equilibrium. **HOWEVER**, this analogy does not mean that the reactants and products are equal in concentration.

For instance, we remove half of the H_2 present in the reaction. As such, the reactants will decrease, and the glasses of water would look like this:

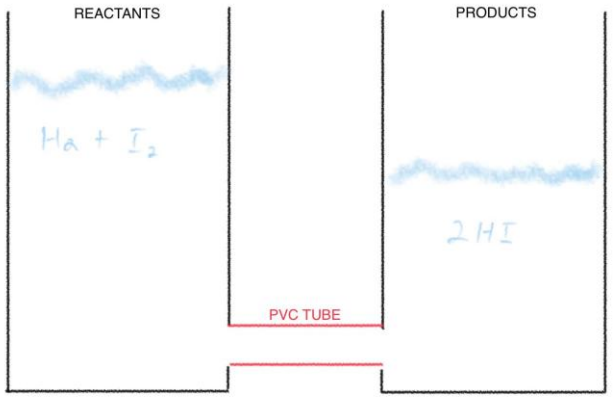


In order to compensate for the disturbance of a change in concentration, **the reaction must shift to the reactant's side.**

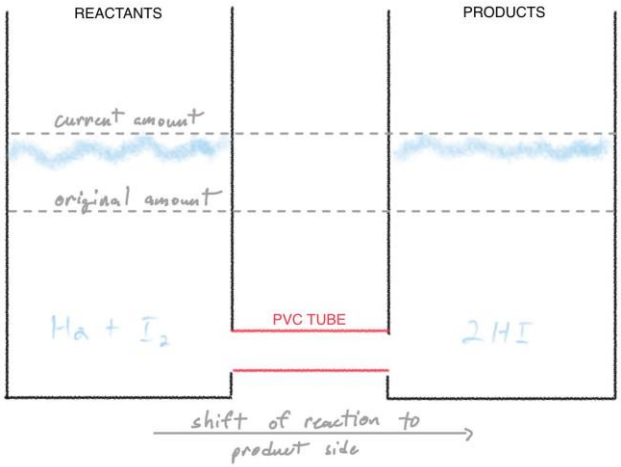


Following the law of conservation of mass, the overall amount of reactants and products will decrease since the amount of reactants decreased originally.

In another example, let's say that the amount of H_2 was increased. The glasses of water, would look like this:



As such, the **reaction must shift to the product side** to achieve equilibrium.m



EFFECT OF CHANGE IN VOLUME AND PRESSURE

A chemical reaction in equilibrium **may be disturbed/stressed by changing the pressure (and likewise volume) of reactants/products**

EFFECT OF CHANGE IN TEMPERATURE

△H>0	△H<0
Endothermic reaction	Exothermic reaction
Heat can be considered as a reactant	Heat can be considered as a product
Adding heat favors the forward reaction	Adding heat favors the reverse reaction
Cooling favors the reverse reaction	Cooling favors the forward reaction

Heating favors away from increase, while cooling favors the decrease.

I. True or False

1. Chemical equilibrium is a static state where all chemical reactions completely stop.
2. In a reversible chemical reaction, the products can react to regenerate the original reactants.
3. At chemical equilibrium, the rate of the forward reaction is equal to the rate of the reverse reaction.
4. The equilibrium constant (K_c or K_P) expression places product concentrations in the numerator and reactant concentrations in the denominator.
5. Pure solids and liquids are included in the equilibrium constant expression regardless of their amounts.
6. A very large equilibrium constant ($K > 1$) indicates that the reaction mixture consists predominantly of products at equilibrium.
7. According to Le Chatelier's Principle, if a stress or disturbance is applied to a system at equilibrium, the system shifts in a direction that minimizes or undoes that stress.
8. The presence of a catalyst alters the position of equilibrium and changes the final equilibrium concentrations.
9. For an exothermic reaction ($\Delta H < 0$), an increase in temperature favors the reverse reaction (shifts to the left).
10. When the pressure on a gaseous equilibrium system is increased by decreasing its volume, the system shifts toward the side with a greater number of moles of gas.

II. Problem Solving

1. A 2.0 L reaction vessel at a certain temperature contains an equilibrium mixture of 2.0 mol N_2, 0.60 mol H_2, and 0.40 mol NH_3 for the Haber process;

   $$N_2\,(g) + 3H_2\,(g) \leftrightarrow 2NH_3\,(g)$$

   Calculate the molar concentrations of each species and determine the value of K_c.
2. Consider a reversible reaction with an equilibrium constant $K_C = 4.5 \times 10^5$. Does this system favor the reactants or the products at equilibrium, and is the equilibrium position located closer to the left or the right?

III. Shift to the Left or Right

Consider the reversible exothermic reaction representing the synthesis of sulfur trioxide:

$$2SO_2\,(g) + O_2\,(g) \leftrightarrow 2SO_3\,(g) \qquad \Delta H < 0$$

1. Addition of O_2 (g) into the reaction mixture
2. Removal of SO_3 (g) from the container
3. Heating the vessel
4. Doubling the volume of the container
5. Introduction of an inert gas at constant volume

ANSWER KEY

Test I.

1. False
2. True
3. True
4. True
5. False
6. True
7. True
8. False
9. True
10. False

Test II.

1. 14.81
2. System favors products at equilibrium ($K_c > 1$)

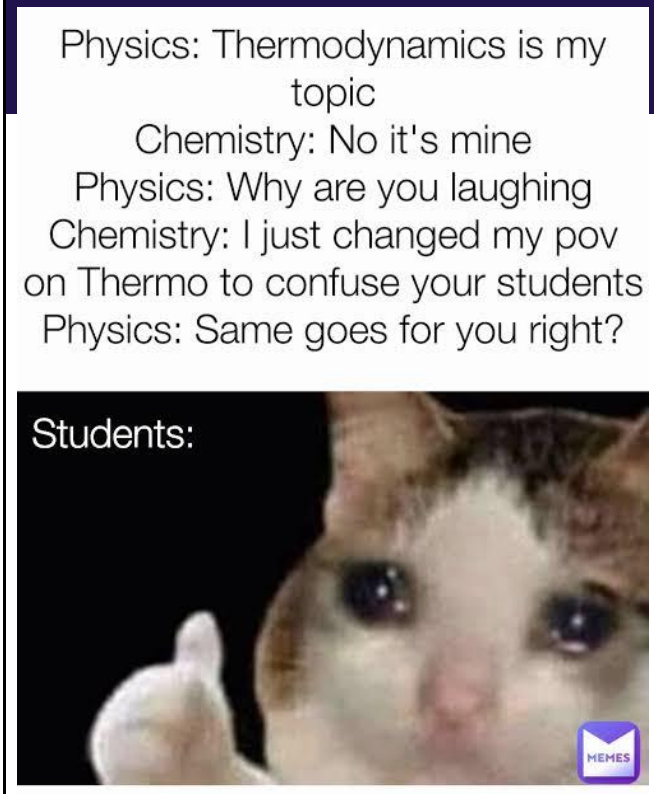
Test III.

1. Shift to products
2. Shift to products
3. Shift to reactants
4. Shift to reactants

5. No effect

CHEMISTRY LEC | UNITS 8

MEME WALL



THERMOCHEMISTRY (UNIT VIII)

Introduction

Thermodynamics

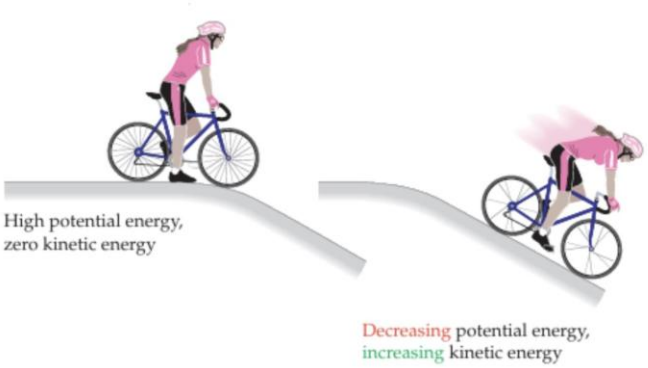
- The science of the relationship between heat and other forms of energy.
- Greek: **therme-**, "heat"; **dynamis**, "power"
- It is the study of energy and its transformations

Thermochemistry

- It is one area of thermodynamics. It concerns the study of the quantity of heat absorbed or evolved by chemical reactions.
- Deals with the study of the relationship between chemical reactions and energy changes that involve heat.

THE NATURE OF ENERGY

- **Energy** is the capacity to do work and transfer heat.
- **Work** is the energy used to cause an object to move against a force.
- **Heat** is the energy used to cause the temperature of an object to increase.



TYPES OF ENERGY

1. **Kinetic Energy** - energy in motion
   - In chemical thermodynamics, the KE of interest is that of atoms and molecules.
2. **Potential Energy** - energy available by virtue of an object's position relative to other objects.
   - "Stored" energy that arises from the attractions or repulsions an object experiences in relation to other objects.
3. **Radiant Energy or Solar Energy** - the energy that comes from the sun.
4. **Chemical Energy** - stored within the structural units of chemical substances.
5. **Thermal Energy** - is the energy associated with the random motion of atoms and molecules.

Modes of Heat Transfer

a. **Conduction or Diffusion** - transfer of energy between objects that are in physical contact.
b. **Convection** - transfer of energy between an object and its environment, due to fluid motion
c. **Radiation** - transfer of energy by the emission of electromagnetic radiation.

UNITS OF ENERGY

- Energy is measured in **two** units.
- **Joule** - is the SI unit for energy - used especially in doing calculations or experiments involving energy, work, or heat.

$$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(2\text{ kg})(1\text{ m/s})^2 = 1\text{ kg-m}^2/\text{s}^2 = 1\text{ J}$$

- 1 J = 1 kg•m²/s²
  - 1000 Joules = 1 kiloJoule

- **Calorie** - another unit of energy - used in food and nutrition measuring the energy in food like calorimetry.
  - The amount of energy required to raise the temperature of 1g of water to 1 degrees Celsius.

1 cal = 4.184 J	1000 cal = 1 kcal = 1 Cal

Sample Conversion Problems: Convert 500 cal to Joules.

$$500\ \cancel{cal} \times \frac{4.184\ J}{1\ \cancel{cal}} = \mathbf{2092\ J}$$

Exercises:

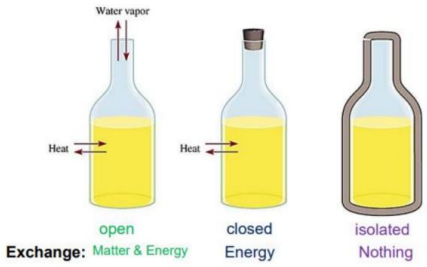
- Example of Kinetic Energy: **A rolling ball**
- Energy stored due to position or arrangement is **Potential energy**.
- A good pitcher can throw a baseball so that it travels between 60 and 90 miles per hour (although speeds in excess of 100 miles per hour have been recorded). A regulation baseball weighing 0.143 kg travels 33.5 meters per second. **What is the kinetic energy of this baseball in joules? In calories?**
  Answer: 80.2 J and 19.2 cal

ENERGY CHANGES IN CHEMICAL REACTIONS



- **System:** The part of the universe chosen for study
- **Surroundings:** everything external to the system
- **System boundary:** interface separating system and surroundings
- **Universe:** combination of system and surroundings.

SYSTEM AND SURROUNDINGS



Systems can be classified as:

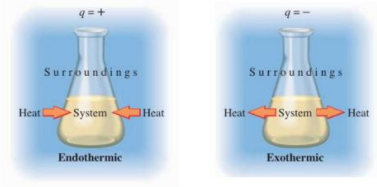
- **Open:** mass and energy can be transferred between system and surroundings
- **Closed:** energy can be transferred but not mass
- **Isolated:** Neither energy nor mass can be transferred between system and surroundings

EXERCISE

Classify each of the following as an open system, a closed system, or an isolated system.

a) Milk kept in a closed thermo flask — Isolated system
b) A student reading in her dorm room. — Open system
c) Air inside a tennis ball. — Closed system

- The **heat of reaction** (at a given temperature) is the value of q required to return a system to the given temperature at the completion of the reaction.



Type of Reaction	*Experimental Effect Noted*	*Result on System*	*Sign of q*
Endothermic	Reaction vessel cools (heat is absorbed)	Energy added	+
Exothermic	Reaction vessel warms (heat is evolved)	Energy subtracted	−

Ebbing, D. D., Gammon, S. D. (2009).

SCHEMATIC OF EXOTHERMIC AND ENDOTHERMIC PROCESS

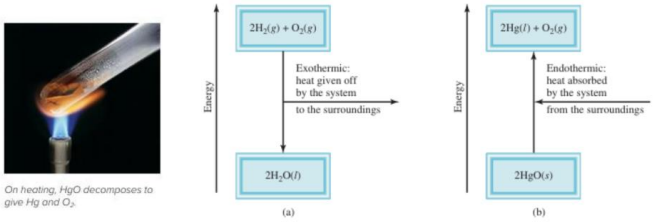


Figure 6.3 (a) An exothermic process. (b) An endothermic process.

Sample Problem 1

Suppose that in an experiment 1 mol of methane burns in oxygen and evolves 890 kJ of heat. Is the reaction endothermic or exothermic? What is the value of q?

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$$

The reaction is exothermic.
There fore the heat of reaction, q, is -890 kJ.

Sample Problem 2

A crystals of barium hydroxide octahydrate, $Ba(OH)_2.8H_2O$, react with crystals of ammonium nitrate, NH_4NO_3, the reaction mixture absorbs 170.8 kJ of heat. Is the reaction endothermic or exothermic? What is the value of q?

$$Ba(OH)_2 \cdot 8H_2O(s) + 2NH_4NO_3(s) \longrightarrow 2NH_3(g) + 10H_2O(l) + Ba(NO_3)_2(aq)$$

The reaction is endothermic.
There fore the heat of reaction, q, is +170.8 kJ.

THERMODYNAMICS

- **State of a system** – the values of all relevant macroscopic properties, for example, composition, energy, temperature, pressure, and volume.
- **Energy, pressure, volume, and temperature are said to be state functions** – properties that are determined by the state of the system, regardless of how that condition was achieved

• $\Delta E = E(final) - E(initial)$

• $\Delta P = P(final) - P(initial)$

• $\Delta V = V(final) - V(initial)$

• $\Delta T = T(final) - T(initial)$

First law of thermodynamics

- Energy cannot be created or destroyed, only changed from one form to another.This means that, whatever energy the system loses, the surroundings gain and vice versa.)



$$\Delta E_{system} + \Delta E_{surroundings} = 0$$

or

$$\Delta E_{system} = -\Delta E_{surroundings}$$

$$C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$$

Exothermic chemical reaction!

Chemical energy lost by combustion = Energy gained by the surroundings
system ***surroundings***

Enthalpy and with the first law of thermodynamics

- Enthalpy (H)- is used to quantify the heat flow into or out of a system in a process that occurs at constant pressure.

$$\Delta E = q + w$$

At constant pressure:

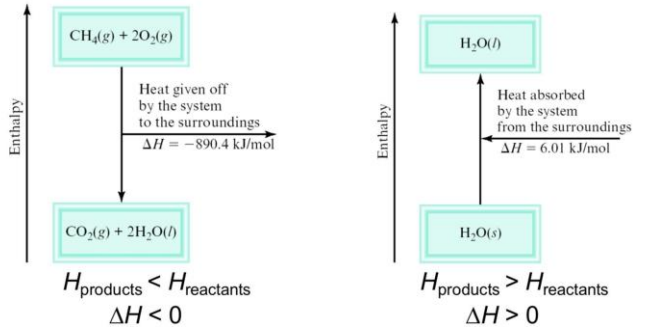
$$q = \Delta H \text{ and } w = -P\Delta V$$

$$\Delta E = \Delta H - P\Delta V$$

$$\Delta H = \Delta E + P\Delta V$$

$$\Delta H = H \text{ (products)} - H \text{ (reactants)}$$

ΔH = heat given off (-) or absorbed (+) during a reaction at constant pressure



Enthalpy change (ΔH) is calculated as products minus reactants. When energy goes down like methane burning, products have less energy than what you started. When heat is released, ΔH is negative, and it's exothermic. When energy goes up like ice melting into water, products have more energy than the start, heat is absorbed, ΔH is positive, and it's endothermic.

❑ Since $\Delta H = H_{final} - H_{initial}$

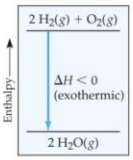
ΔH = H(products) - H(reactants)

The enthalpy change that accompanies a reaction is called the enthalpy of reaction or the heat of reaction ΔH_{rxn}

EXAMPLE: Combustion of Hydrogen

$$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)} \quad \Delta H = -483.6 \text{ kJ}$$

- *The reaction occurs under constant pressure, and the negative sign tells us its exothermic*
- *Balanced chemical equations of this sort are Called thermo-chemical equations*



I. Identify the term being described.

1. The science of the relationship between heat and other forms of energy.
2. The area of thermodynamics that studies the quantity of heat absorbed or evolved by chemical reactions.
3. The capacity to do work and transfer heat.
4. The energy used to cause an object to move against a force.
5. The energy used to cause the temperature of an object to increase.
6. Energy in motion.
7. Energy available by virtue of an object's position relative to other objects.
8. The energy that comes from the sun.
9. Energy stored within the structural units of chemical substances.
10. The energy associated with the random motion of atoms and molecules.

II. Identify whether each description refers to Conduction, Convection, or Radiation.

1. Transfer of energy between objects that are in physical contact.
2. Transfer of energy between an object and its environment due to fluid motion.
3. Transfer of energy by the emission of electromagnetic radiation.

III.

1. Enumerate the two units of energy discussed in this lesson.
2. Enumerate the three classifications of systems based on how they exchange energy and mass with their surroundings.
3. Enumerate the four state functions mentioned in the lesson.
4. Enumerate the five types of energy discussed under "Types of Energy."
5. Enumerate the three modes of heat transfer.

IV.

1. A regulation baseball weighing 0.143 kg travels at 33.5 meters per second. What is the kinetic energy of this baseball in joules? In calories?
2. Convert 350 joules to calories. (1 cal = 4.184 J)
3. A 0.50 kg object is moving at 10.0 m/s. Calculate its kinetic energy in joules, then convert your answer to calories.
4. A chemical reaction absorbs 45 kJ of heat from its surroundings. Is this reaction exothermic or endothermic? What is the sign of ΔH? Explain your answer.

Part I: 1. Thermodynamics 2. Thermochemistry 3. Energy 4. Work 5. Heat 6. Kinetic Energy 7. Potential Energy 8. Radiant/Solar Energy 9. Chemical Energy 10. Thermal Energy

Part II: 1. Conduction (Diffusion) 2. Convection 3. Radiation

Part III:

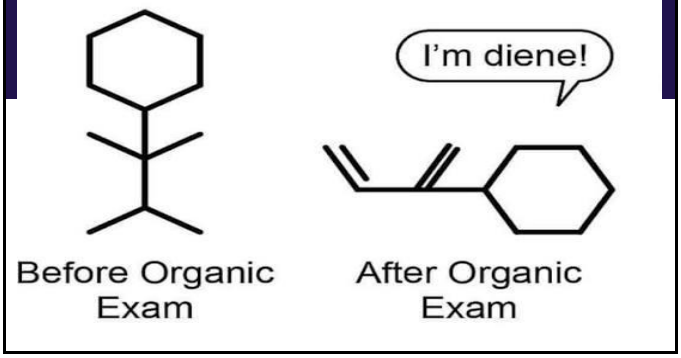
1. Joule, Calorie
2. Open, Closed, Isolated
3. Energy (E), Pressure (P), Volume (V), Temperature (T)
4. Kinetic, Potential, Radiant/Solar, Chemical, Thermal
5. Conduction, Convection, Radiation

Part IV.

1. KE = 80.2 J = 19.2 cal 25. 350 J ÷ 4.184 J/cal = 83.65 cal
2. KE = ½mv² = ½(0.50 kg)(10.0 m/s)² = 25 J = 5.97 cal
3. Endothermic; ΔH is positive, since heat is absorbed from the surroundings

CHEMISTRY LEC | UNIT 10

MEME WALL



ORGANIC CHEMISTRY

- **Organic chemistry:** study of compounds made of carbon
- **Organic compounds** - made up of carbon and only few other elements
  - Chief: H, O, N (aka **CHON**)
  - Also present: **halogens**
  - **Does not include ionic compounds and metal carbonate salts**
- Studies structure of carbons and reactions

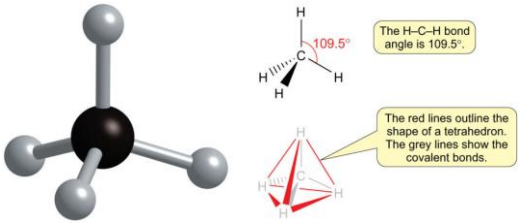
HISTORY OF ORGANIC CHEMISTRY

- **VITAL FORCE THEORY**
  - A "vital force" in living organisms was necessary to produce an organic compound
  - Proponent: Berzelius
- **Friedrich Wohler** - found 1 organic salt and urea (OC) after a chemical reaction involving inorganic compounds
  - Ended vital force theory
  - Chemists start to synthesize OCs
- **85%** of all known compounds are organic
- OrgChem serves as basis of biochemistry (biochemicals)

ORGANIC VS INORGANIC COMPOUNDS

	Organic (butane)	**Inorganic (sodium chloride)**
Bonding	Covalent	Ionic
State @ room temp & atmospheric pressure	Gas	Solid (crystal lattice)
Boiling & melting point	Low (weak IMF)	High (strong IMF)
Solubility in H2O (polar solvent)	insoluble	High (polar)
Solubility in organic solvents	High	insoluble
Flammability	Flammable (Combustion)	Nonflammable
Electrical conductivity	Nonconductor (nonelectrolytes)	Conducts in solution & molten

PROPERTIES OF CARBON



1. **Catenation-** forms stable bonds with other C atoms (Self-linking)
2. Forms stable bonds with other elements (SON & halogens)
   a. Presence of the other elements confers many new physical and chemical properties
3. **Versatility** - Flexible ; can do single, double, or triple bonds to produce various structures with different properties
   a. Can form up to 4 bonds
4. Many ways to arrange (nearly limitless)
   a. Branch
   b. Ring
   c. Linear

ORGANIC STRUCTURE

General guidelines for bonds in organic molecules:

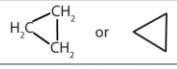
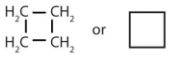
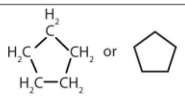
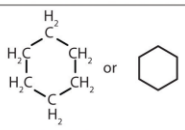
Elements in uncharged	Covalent bonds	Unshared pairs
Carbon	4	0
Hydrogen	1	0
Nitrogen	3	1
Oxygen	2	2
Halogens	1	3

It is especially important to take note of C and H, as it can help you infer expanded structure from skeletal

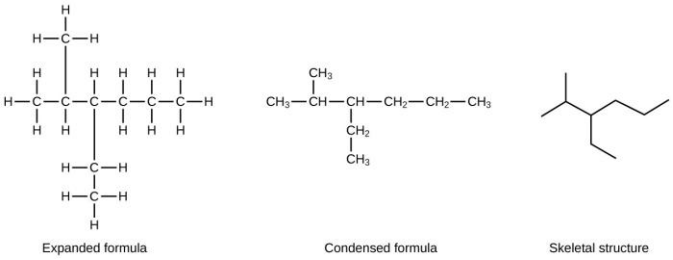
- **STRUCTURAL FORMULA** - shows atoms present and bonds that connect them

- VSEPR model - geometry based on electron repulsion

1. **EXPANDED STRUCTURE**: all bonds and elements are shown
    - Lewis w/o electron dots
2. **CONDENSED STRUCTURE** - bonds are implied rather than fully shown

Name	Molecular Formula	Structural Formula
cyclopropane	C_3H_6	
cyclobutane	C_4H_8	
cyclopentane	C_5H_{10}	
cyclohexane	C_6H_{12}	

3. **SKELETAL / LINE-ANGLE FORMULA** - line represents carbon-carbon bond (hides C and H)
    a. Vertices and line terminus represent carbon atom
    b. Must have bends to show distinction between carbons
    c. H bonds are deduced by how many bonds are on C
    d. Still shows double and triple bonds



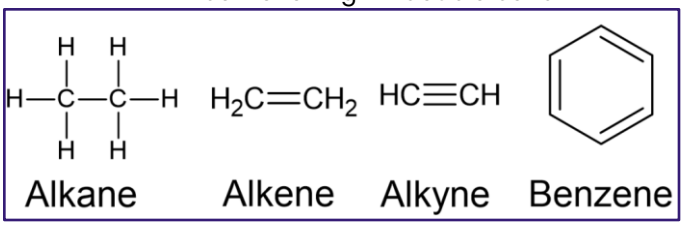
Expanded formula | Condensed formula | Skeletal structure

FUNCTIONAL GROUPS

- A group of atoms within a molecule that shows a characteristic set of predictable physical and chemical properties (group based on similar properties).
- Importance:
    - Undergo same types of chemical reactions no matter what organic molecule they are found in
    - Determine physical and chemical properties of a molecule
    - Units by which we divide organic compounds into families
    - Provide basis of naming organic compounds

HYDROCARBONS

1. **Alkane** - single bond
2. **Alkene** - has at least 1 double bond
3. **Alkyne** - at least 1 triple bond
4. **Aromatic compound** - ring form(phenyl group)
    - benzene ring - 2 double bond



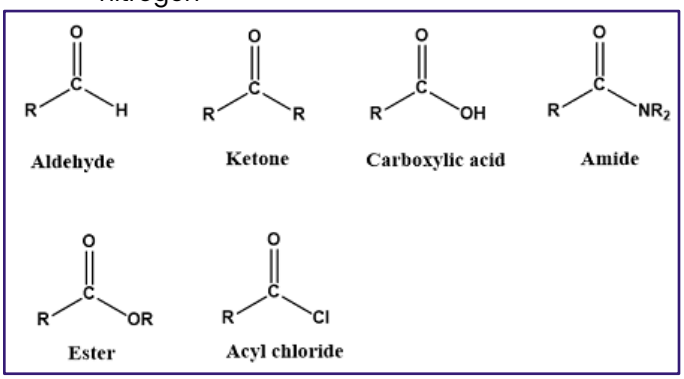
c-Z; z is more electronegative (SON)

- Single bonds

1. **Alkyl halide** (halogens) halo group - Br, Cl, I, F
2. **Alcohol (Hydroxyl)** R-O-H
3. **Ether (Alkoxy)** - R-O-R (R groups on EITHER side of O)
4. **Amine** - C-N amino
5. **Thiol**- sulfhydryl grp, mercapto

CARBONYL C=O

1. **Aldehyde** - has H and O bonded to C separately
2. **Ketone** - C=O carbonyl (C is the one attached to R groups (R-C-R))
3. **Carboxylic acid** - COOH
4. **Ester** - single bond with oxygen + carbonyl
5. **Amide** - double bonded carbon (carbonyl) with nitrogen



Isomers

- Compounds that have the **same formula** but **different structures**
- Due to these **structural differences,** they differ in terms of physical and chemical properties
- There are two types of **isomers:**

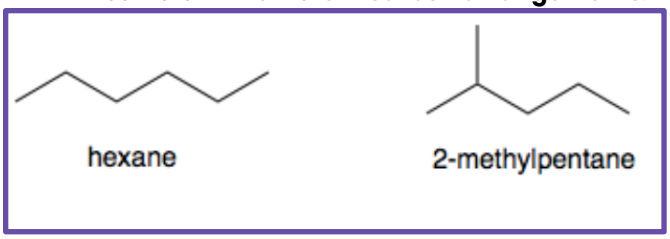
- Constitutional Isomers
- Steroisomers

Constitutional Isomers

- Molecules have the **same formula** but have a **different connectivity** of atoms
- Three types of constitutional isomers:
  - Skeletal Isomers
  - Positional Isomers
  - Functional Group Isomers

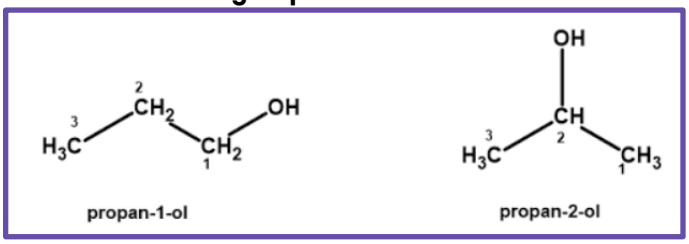
Skeletal Isomers

- Isomers with **different carbon arrangements**.



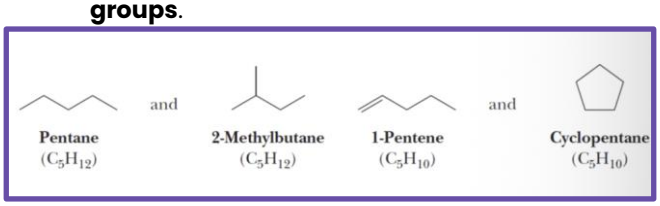
Positional Isomers

- Isomers that **differ** in the **location of the functional group**.



Functional Group Isomers

- Isomers that contain **different functional groups**.



(1) Pentane and 2-Methylbutane

Pentane (C_5H_{12}) The 5 carbon atoms are arranged in a **straight chain**.

2-Methylbutane (C_5H_{12}) The same 5 carbon atoms are arranged in a **branched chain**.

They have the **same molecular formula (C_5H_{12})**, but the carbon atoms are connected differently. Therefore, they are **skeletal isomers**.

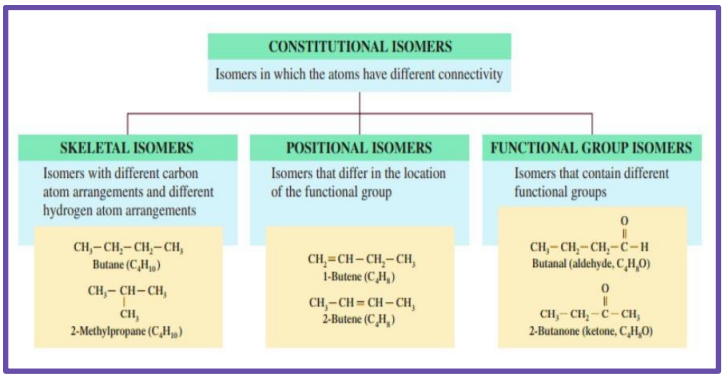
(2) 1-Pentene and Cyclopentane

1-Pentene (C_5H_{10}) The 5 carbon atoms form an **open chain** with a double bond.

Cyclopentane (C_5H_{10}) The 5 carbon atoms form a **ring**.

They have the **same molecular formula (C_5H_{10})**, but their atoms are connected differently. One is a **chain**, while the other is a **ring**, so they are also **constitutional isomers**.

Among the kinds of constitutional isomers, we've seen are skeletal, functional, and positional isomers.



Skeletal isomers have different arrangements of their carbon atoms.

For example, butane is a straight chain, while 2-methylpropane is branched.

Think: the carbon skeleton changes.

Positional isomers have the same carbon skeleton, but the **position of a double bond or functional group changes**.

For example, 1-butene and 2-butene both have a double bond, but it is in a different location.

Think: the position changes.

Functional group isomers have the same molecular formula but **different functional groups**.

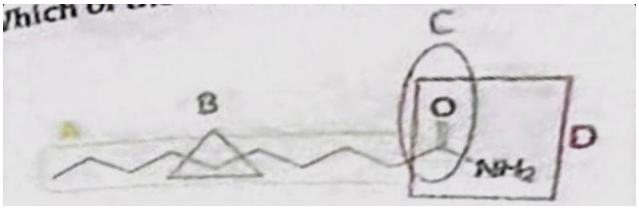
For example, butanal has an aldehyde group, while 2-butanone has a ketone group.

Think: the functional group changes

----------END OF CHAPTER----------

Q&A FROM LAST QUIZ

1. **Who discarded the vitalism theory?**
    a. Friedrich Wöhler (1828)
2. **What compound produced the 1st synthetic organic compound?**
    a. Urea
3. **2 important breakthroughs ni Wohler**
    a. Disproved vital force theory
    b. Organic compounds can be made from inorganic substances
4. **2 sources of organic compounds**
    a. Plant and animal matter
    b. synthetic compounds
5. **Identify the functional groups**



    a. alkane chain
    b. Uhhh ;P hehehehehehehehee mas kinikilig ka pa sa'kin pag tinititigan na sayong mga mata
    c. carbonyl
    d. amide
6. **What hybridization is found in alkenes?**
    a. sp^2
7. **Hybridization happens when ___.**
    a. Orbitals combine to form new orbitals EUHHH basta yun
8. **What orbitals join in hybridization?**
    a. S and P orbitals

MOCK TEST

1. According to VSEPR theory, what bond angle is formed around a tetrahedral carbon atom with four single bonds?
    A) 90°
    B) 109.5°
    C) 120°
    D) 180°

2. What orbital hybridization is found in double-bonded carbons of alkenes?
    A) sp^3 hybridization
    B) sp^2 hybridization
    C) sp hybridization

3. Which functional group family features an oxygen atom single-bonded between two carbon atoms?
    A) Alcohol
    B) Ether
    C) Ester
    D) Ketone

4. Which functional group contains a carbonyl carbon attached directly to an alkoxy group?
    A) Ester
    B) Amide
    C) Aldehyde
    D) Carboxylic acid

5. How does an aldehyde structurally differ from a ketone?
    A) Aldehydes contain nitrogen, while ketones contain sulfur.
    B) Aldehydes contain a carbonyl carbon attached to at least one hydrogen atom, while ketones have a carbonyl carbon attached to two carbon groups.
    C) Aldehydes have double-bonded oxygen, while ketones have single-bonded oxygen.
    D) Aldehydes are always cyclic, while ketones are always open-chain.

6. Compared to inorganic ionic salts like sodium chloride, typical organic compounds like butane display which set of physical properties?
    A) High melting points and nonflammability
    B) Ionic bonding and high water solubility
    C) Covalent bonding, flammability, and low melting points
    D) Electrical conductivity in aqueous solutions

7. How many covalent bonds and unshared electron pairs does a neutral carbon atom form in organic compounds?
    A) 4 covalent bonds and 0 unshared electron pairs
    B) 4 covalent bonds and 2 unshared electron pairs
    C) 3 covalent bonds and 1 unshared electron pair
    D) 2 covalent bonds and 3 unshared electron pairs

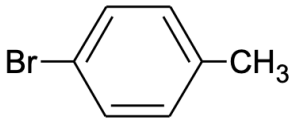
8. Which functional group family features a carbon atom single-bonded to a halogen atom (such as fluorine, chlorine, bromine, or iodine)?
    A) Thiol
    B) Alkyl halide
    C) Alcohol
    D) Amine

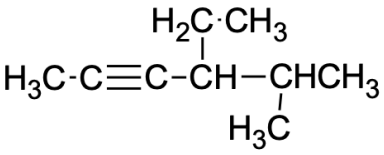
9. Which unique bonding capability of carbon accounts for the vast diversity of over 10 million organic structures?
    A) Its ability to form up to 4 stable covalent bonds with other carbon atoms in linear chains, branched chains, and rings
    B) Its tendency to lose 4 valence electrons to form ionic lattices
    C) Its high solubility as a metal in inorganic solvents
    D) Its inability to form double or triple bonds with nonmetals

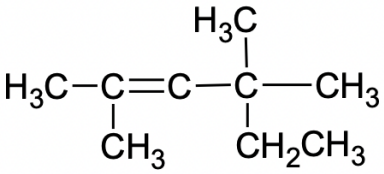
10. In the compound acetamide (CH_3CONH_2), a carbonyl carbon is directly bound to an amino-derived nitrogen group. Which functional group family does this structure belong to?
    A) Amine
    B) Ester

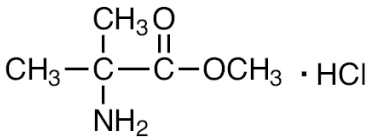
C) Amide
D) Anhydride

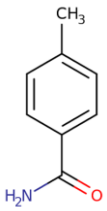
Name the following compounds:

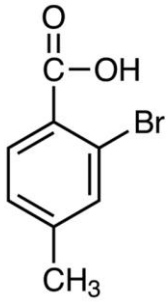
1. 

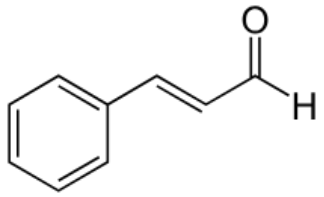
2. 

3. 

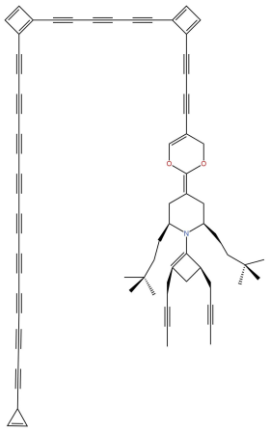
4. 

5. 

6. 

7. 

BONUS:



ANS KEY:

1. B
2. B
3. B
4. A
5. B
6. C
7. A
8. B
9. A
10. C

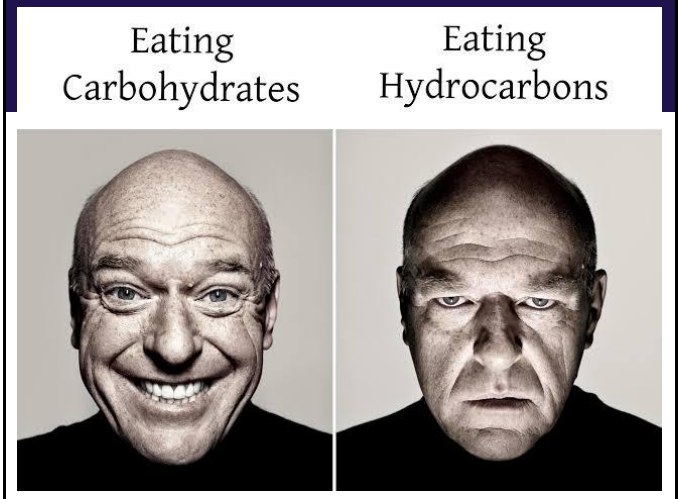
1. P-bromotoluene
2. 4-ethyl-5-methyl-2-hexyne

3. 2,4,4-trimethyl-2-hexene
4. 2-Amino-2-methylpropanoate Hydrochloride
5. 4-Methylbenzamide
6. 2-Bromo-4-methylbenzoic Acid
7. 3-phenylprop-2-en-1-al

Bonus: aq auhhauahuhahahahahahha ahahahahahah ahaha ah haha hahahaha 1,2,3,4,4,4,4,9,9,69,108-epoxymethylethylpentylhexylhappyholihappydiwalihappybirthday butanpentangoodnight-2,3,6,9,69-mutanone DAW kuno

CHEMISTRY LEC I UNIT 11

MEME WALL



HYDROCARBONS

- **Hydrocarbon** - A compound composed only of carbon and hydrogen.
- 2 classes of hydrocarbons:
    - **Saturated** - contain only <u>single</u> covalent bonds
    - **Unsaturated** - contain at least one <u>double</u> or <u>triple</u> bond

ALKANES

- **Alkanes** - Hydrocarbons that contain only carbon-carbon single bonds.
- There are 2 parts of the IUPAC name of an alkane with an <u>unbranched chain</u>:
    - Prefix showing number of carbon atoms in the chain
    - -ane: shows that the compound is a saturated hydrocarbon

# of carbon atoms	Prefix
1	meth-
2	eth-
3	prop-
4	but-
5+	Greek numbers

- Name of an alkane with a <u>branched</u> chain:
    - **Parent name** - longest chain of carbon atoms (follows previous nomenclature)
    - **Substituent names** - branches/groups bonded to the parent chain
        - **Alkyl group** - a substituent group derived from an alkane by removal of a hydrogen atom
        - Commonly represented by symbol R-
        - Replace -ane with -yl

<u>SOURCES OF ALKANES</u>

- **Natural gas**
    - 90-95% methane
    - 5-10% ethane
    - Propane, butane, isobutane
- **Petroleum** - A thick, viscous liquid mixture of thousands of compounds, most of them hydrocarbons formed from the decomposition of marine plants and animals.

<u>PHYSICAL PROPERTIES</u>

- **Almost complete lack of polarity**
    - Electronegativity differenced between C and H is 0.4 (NONPOLAR) on the Pauling scale!
- **Melting & boiling points**
    - Lower than those of almost any other type of compound of the same molecular weight
- **Constitutional isomer**
    - Constitutional isomers are different compounds and have different physical and chemical properties
- **Solubility**
    - Not soluble in water (no H bonds)
    - Liquid alkanes are soluble in each other
    - Soluble in other nonpolar organic compounds
- **Density**
    - 0.7 g/mL on average, 0.8 for higher molecular weight alkanes

<u>REACTIONS OF ALKANES</u>

- **Oxidation (combustion)**
    - Basis for their use as energy sources for heat and power
- **halogenation**
    - Reaction with halogens = substitution reaction

IUPAC NOMENCLATURE RULES

1. The name for an alkane with an unbranched chain of carbon atoms consists of a prefix showing the number of carbon atoms and the ending -ane.
2. For branched-chain alkanes, the longest chain of carbon atoms is the parent chain and its name is the root name.
3. Give each substituent on the parent chain a name and a number. Use a hyphen to connect the number to the name.
4. If there is one substituent, number the parent chain from the end that gives the substituent the lower number.
5. If the same substituent occurs more than once:
    a. Number the parent chain from the end that gives the lower number to the substituent encountered first.
    b. Indicate the number of times the substituent occurs by a prefix di-, tri-, tetra-, penta-, hexa-, and so on.
    c. Use a comma to separate position numbers.
6. If there are two or more different substituents:
    a. List them in alphabetical order.
    b. Number the chain from the end that gives the lower number to the substituent encountered first.
    c. If there are different substituents at equivalent positions on opposite ends of the parent chain, give the substituent of lower alphabetical order the lower number.
    d. Do not include the prefixes di-, tri-, tetra-, and so on or the hyphenated prefixes sec- and tert- in alphabetizing (only alphabetize the ethyl and methyl parts (root))

COMMON NAMES

- Number of carbon atoms determines the name (regardless of branching)
- Iso shows that one end of an otherwise unbranched chain terminates in (CH3)2CH-
- IUPAC for complex alkanes

CYCLOALKANES

- **Cyclic hydrocarbon** - A hydrocarbon that contains carbon atoms joined to form a ring.
- **Cycloalkane** - A cyclic hydrocarbon in which all carbons of the ring are saturated (have only carbon-carbon single bonds).

NOMENCLATURE

- Open-chain alkane: cyclo-, name each substituent
- 1 substituent -> no need for location number
- 2 substituents -> number alphabetically

ALKENES & ALKYNES

- **Alkene**: A hydrocarbon that contains one or more carbon-carbon double bonds.
    - **Ethylene** is the simplest alkene.
    - Polyenes - contain several double **bonds**
- **Alkyne**: A hydrocarbon that contains one or more carbon-carbon triple bonds.
    - **Acetylene** is the simplest alkyne.

STRUCTURE

- VSEPR model: Bond angles of ~120
    - Angles may be greater than 120 in substituted alkenes
- Cis-trans isomerism
    - Restricted rotation about a C=C, an alkene with 2 diff groups on each C of the double bond shows cis-trans isomerism

PHYSICAL PROPERTIES

❑ Alkenes and alkynes are nonpolar compounds.
❑ The only attractive forces between their molecules are London dispersion forces.
❑ Their physical properties are similar to those of alkanes with the same carbon skeletons.
❑ Alkenes and alkynes are insoluble in water but soluble in one another and in nonpolar organic liquids.
❑ Alkenes and alkynes that are liquid or solid at room temperature have densities less than 1.0 g/mL; they float on water.

IUPAC NAMES OF ALKENES & ALKYNES

1. Parent name - longest chain containing double or triple bond
2. Number chain from end giving lower numbers to the double or triple bond
3. Locate double or triple bond by number of its first carbon
4. -ene show presence of the double bond, -yne shows presence of triple bond

5. Branched chain alkenes and alkynes are named similarly to branched alkanes

COMMON NAMES

- Used for some alkenes and alkynes with low molecular weight (iso)

CYCLOALKENES

- Number the carbon atoms of the ring double bond 1 and 2 in the direction that gives the lower number to the substituent encountered first.
- not necessary to explicitly number the position of the double bond in a cycloalkene as in linear alkenes.
- Number and list substituents in alphabetical order.

POLYENES

- Alkenes that contain more than one double bond are named as **alkadienes, alkatrienes**, etc.

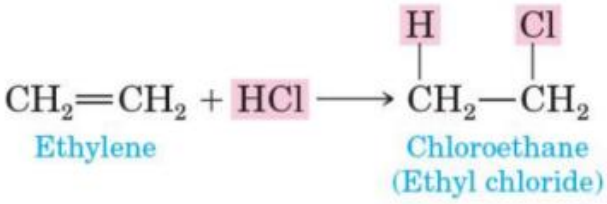
REACTIONS OF ALKENES

GENERAL PROPERTIES

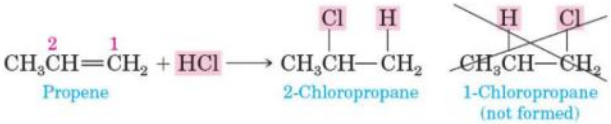
- Most alkene addition reactions are exothermic (products are more stable than reactants)
- Does not mean that alkene addition reactions occur rapidly
    - Depends on activation energy
    - Many require a catalyst

ADDITION OF HX

- Addition of HX (HCl, HBr, HI) to an alkene gives a **haloalkane**
    - H adds to one carbon of the C=C and X to the other



    - **Regioselective** - preference of a reaction to form/break bonds at one specific part of a molecule over other parts
    - **Markovnikov's rule** - H adds to the less substituted carbon and X to the more substituted carbon



ADDITION OF HCL TO 2-BUTENE

1. Add a proton
2. Reaction of an electrophile and a nucleophile to form a new C-C bond

ADDITION OF H2O

- Hydration - addition of water
- Acid catalyzed, most commonly by H2SO4
- Follows Markovnikov's rule
    - OH adds to the more substituted carbon

ADDITION OF CL2 BR2

- Takes place readily at room temperature
- Generally carried out using pure reagents or mixing them in a nonreactive organic solvent
- Addition of Br2 is a useful qualitative test for presence of a C=C
- Br2 has a deep red color, dibromoalkanes are colorless

ADDITION OF H2 - REDUCTION

- Virtually all alkenes add H2 in the presence of a transition metal catalyst, commonly Pd, Pt, or Ni.

POLYMERIZATION

- Most important reaction of alkenes
    - Polymer: long-chain molecule synthesized by bonding together monomers

----------END OF CHAPTER----------

Alkene Reactions Multiple-Choice Quiz, choose the best answer for each question.

1. Most alkene addition reactions are:
A) endothermic B) exothermic C) always fast D) require no catalyst

2. Adding HX (like HCl) to an alkene produces:
A) alkane B) haloalkane C) alcohol D) polymer

3. According to Markovnikov's rule, H adds to which carbon of the C=C bond?
A) more substituted carbon B) less substituted carbon C) either equally D) the shorter carbon chain

4. Hydration of alkenes means adding which molecule?
A) H_2 B) H_2O C) Cl_2 D) HBr

5. What catalyst is commonly used for alkene hydration?
A) Pd metal B) H_2SO_4 C) Pt D) Ni

6. When Br_2 reacts with an alkene, the reddish-brown Br_2 color:
A) stays the same B) disappears C) becomes blue D) becomes yellow

7. Addition of H_2 to an alkene is also called:
A) hydration B) reduction C) halogenation D) polymerization

8. Which of the following is a common catalyst for adding H_2 to alkenes?
A) H_2SO_4 B) HCl C) Pd D) Br_2

9. The Br_2 test is used to detect the presence of:
A) C-C single bond B) C=C double bond C)-OH group D) haloalkane

10. Polymerization of alkenes makes:
A) small monomers B) long-chain molecules C) alcohols D) haloalkanes

ANS.KEY: everything is B except 8.C

CHEMISTRY LEC | UNITS 6 & 7

TOPIC: CHEMICAL EQUILIBRIUM

MID-TERM | S.Y. 2026-2027 | RANAY & ORBITA

CHEMICAL EQUILIBRIUM II

EQUILIBRIUM-CONSTANT EXPRESSIONS

- The **influence** of **concentration or pressure** on the **position of a chemical equilibrium** in quantitative terms.
- Allows us to predict the **direction (forward/backward) and completeness** of chemical reactions.
- Yields **NO INFORMATION** concerning the rate of a reaction
- Some reactions have highly favorable equilibrium constants but are of little analytical use because they are slow. The use of a **catalyst** can overcome this limitation.
    - A catalyst will NOT shift the equilibrium. Instead, it only speeds up the reaction.
- The equilibrium-constant expression becomes:

$$wW + xX \leftrightarrow yY + zZ$$

$$K = \frac{[Y]^y[Z]^z}{[W]^w[X]^x}$$

$$K = \frac{[Y]^y}{[W]^w[X]^x}$$

- **Note that the square-bracket terms are:**

1. Molar concentrations if they represent dissolved solutes.
2. Partial pressures in atmospheres if they are gas-phase reactants or products. [Z]^z is replaced with pz (partial pressure of Z in the atmosphere.

Note: **No term for Z** is included in the equation if this species is a **pure solid, pure liquid, or solvent** of a dilute solution.

- **Constant K** is a temperature-dependent numerical quantity called the equilibrium constant.
- By convention, the concentrations of the products in the written equation are always placed in the **numerator**. The concentrations of the reactants are always in the **denominator**.

$$K = \frac{a_Y^y a_Z^z}{a_W^w a_X^x}$$

where a_Y, a_Z, a_W, and a_X are the activities of species Y, Z, W, and X.

TYPES OF EQUILIBRIUM CONSTANTS IN ANALYTICAL CHEMISTRY

TABLE 9-2
Equilibria and Equilibrium Constants Important in Analytical Chemistry

Type of Equilibrium	Name and Symbol of Equilibrium-Constant	Typical Example	Equilibrium-Constant Expression
Dissociation of water	Ion-product constant, K_w	$2H_2O \rightleftharpoons H_3O^+ + OH^-$	$K_w = [H_3O^+][OH^-]$
Heterogeneous equilibrium between a slightly soluble substance and its ions in a saturated solution	Solubility product, K_{sp}	$BaSO_4(s) \rightleftharpoons Ba^{2+} + SO_4^{2-}$	$K_{sp} = [Ba^{2+}][SO_4^{2-}]$
Dissociation of a weak acid or base	Dissociation constant, K_a or K_b	$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$	$K_a = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]}$
		$CH_3COO^- + H_2O \rightleftharpoons OH^- + CH_3COOH$	$K_b = \frac{[OH^-][CH_3COOH]}{[CH_3COO^-]}$
Formation of a complex ion	Formation constant, β_n	$Ni^{2+} + 4CN^- \rightleftharpoons Ni(CN)_4^{2-}$	$\beta_4 = \frac{[Ni(CN)_4^{2-}]}{[Ni^{2+}][CN^-]^4}$
Oxidation/reduction equilibrium	K_{redox}	$MnO_4^- + 5Fe^{2+} + 8H^+ \rightleftharpoons Mn^{2+} + 5Fe^{3+} + 4H_2O$	$K_{redox} = \frac{[Mn^{2+}][Fe^{3+}]^5}{[MnO_4^-][Fe^{2+}]^5[H^+]^8}$
Distribution equilibrium for a solute between immiscible solvents	K_d	$I_2(aq) \rightleftharpoons I_2(org)$	$K_d = \frac{[I_2]_{org}}{[I_2]_{aq}}$

A. THE ION PRODUCT CONSTANT OF WATER Kw

- Water is autoionization/self-ionization. It dissociates to hydrogen ion and hydroxide ion

$H_2O\ (l) \rightleftharpoons H^+\ (aq) + OH^-\ (aq)$ $\quad K_c = \frac{[H^+][OH^-]}{[H_2O]}$ $\quad [H_2O] = \text{constant} = 1$

Therefore, $K_c[H_2O] = \boldsymbol{K_w = [H^+][OH^-]}$

The ion product of water (K_w) is the product of the molar concentrations of H^+ and OH^- ions **at a particular temperature**

- **AT 25° C**

At 25°C
$K_w = [H^+][OH^-] = 1.00 \times 10^{-14}$

	Solution Is
$[H^+] = [OH^-]$	neutral
$[H^+] > [OH^-]$	acid
$[H^+] < [OH^-]$	base

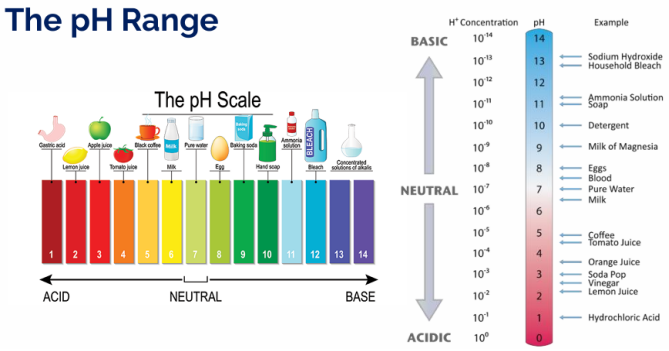
$[H_3O^+] = [OH^-] = 1.0 \times 10^{-7}$

EXAMPLE: The concentration of OH ions in a certain household ammonia cleaning solution is 0.0025 M. Calculate the concentration of H ions at 25° C.

Recall that $K_w = [H^+][OH^-]$, we will rearrange the equation

$$[H^+] = \frac{K_w}{[OH^-]} = \frac{1.00 \times 10{-}14}{0.0025\ M} = 4.0 \times 10^{-12}\ M$$

pH RANGE



pOH and pH

- **pH** - the measure of hydronium ion concentration
- Higher pH = less acidic, more basic.
- Why:

Formulas to find pH and pOH:

$pH = -\log [H_3O^+]$	$[H_3O^+] = 1.0 \times 10^{-14} / [OH^-]$
$pOH = -\log [OH^-]$	$[OH^-] = 1.0 \times 10^{-14} / [H_3O^+]$
$pH + pOH = 14$	$H_3O^+ = \text{antilog}\ (-pH)$
$[H_3O^+]\ [OH^-] = 1.0 \times 10^{-14}$	$OH^- = \text{antilog}\ (-pOH)$

ACID & BASE DISSOCIATION CONSTANTS

- **Partial dissociation** occurs when a weak acid or weak base is dissolved in water.
- $K_w = K_a\ K_b$

CONCEPTUAL DEFINITION OF ACIDS AND BASES

	Acid	**Base**
Arrhenius	Produces H+ ions in water Produced thru ionization	Produces OH- ions in water Produced thru dissociation
Brønsted-Lowry	Proton (H+) donor	Proton (H+) acceptor
Lewis	Electrophile; accepts electron pair	Nucleophile; donates an electron pair

More on Brønsted-Lowry definitions:

- BL acid and BL base production must occur **simultaneously.**
- **Amphiphrotic substance** - can either lose or accept a proton, can function as a BL acid or a BL base
  - Structural requirement: presence of both a hydrogen atom and a lone pair of electrons.
  - Ex. H_2O
- **Conjugate acid-base pair** - any pair of molecules/ions that can be interconverted by proton transfer
  - When an acid transfers a proton to a base, the acid is converted to its conjugate base
  - When a base accepts a proton, it is converted to its conjugate acid.
  - Ex. Ammonia $[NH_3]$ (base) and Ammonium ion $[NH_4^+]$ (conjugate acid)

COMMON ION EFFECT

- **Common ion effect** - shift in equilibrium caused by addition of a compound having an ion in common with the dissolved substance
    - Its presence decreases the ionization of a weak acid or a weak base.
- **Buffer solution** - solution containing both:
    - A weak acid or weak base
    - Salt of the weak acid or weak base
- Buffers tend to resist change in pH when small to moderate amounts of a strong acid or strong base are added.
    - It contains both an acid to neutralize OH- ions and a base to neutralize H+ ions.
    - Must not consume each other through a neutralization reaction
- **Henderson-Hasselbalch equation** - used to calculate the pH of buffer solutions
    - Obtained by expressing each term in the form of its negative logarithm and inverting the concentration to keep all signs positive.

Henderson-Hasselbalch equation

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

***EXAMPLE*:** To mimic blood buffer, a scientist prepared a 1.00L buffer containing 0.0025 mol carbonic acid and 0.025 mol hydrogen carbonate ion. Calculate the pH of a buffer. The Ka of carbonic acid is 4.3 X 10 -7?

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$

$$pH = -\log K_a + \log \frac{[A^-]}{[HA]}$$

Use the **Henderson–Hasselbalch equation**. This equation is used to calculate the pH of a buffer when you know the weak acid, its conjugate base, and its Ka

$$pH = -\log (4.3 \times 10^{-7}) + \log \frac{(0.025)}{(0.0025)} = -\log (4.3 \times 10^{-7}) + \log (10)$$

Replace the numbers in the Henderson–Hasselbalch equation with the values given in the problem. **Ka =4.3×10-7 , [A-]=0.025 , and [HA]=0.0025**

$$pH = 6.37 + \log (10) = 6.37 + 1.00$$

$$pH = 7.37$$

EXAMPLE 2: Calculate the pH of a buffer that is 0.200 M in NH3 and 0.300 M in NH4Cl. The Kb is 1.8 x 10-5

$$NH_3 + H_2O \leftrightarrow NH_4^+ + OH^-$$

Solution

In Appendix 3, we find that the acid dissociation constant K_a for NH_4^+ is 5.70×10^{-10}. The equilibria we must consider are

$$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+ \qquad K_a = 5.70 \times 10^{-10}$$

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^- \qquad K_b = \frac{K_w}{K_a} = \frac{1.00 \times 10^{-14}}{5.70 \times 10^{-10}} = 1.75 \times 10^{-5}$$

Using the arguments that led to Equations 9-25 and 9-26, we find that

$$[NH_4^+] = c_{NH_4Cl} + [OH^-] - [H_3O^+] \approx c_{NH_4Cl} + [OH^-]$$

$$[NH_3] = c_{NH_3} + [H_3O^+] - [OH^-] \approx c_{NH_3} - [OH^-]$$

Because K_b is several orders of magnitude larger than K_a, we have assumed that the solution is basic and that $[OH^-]$ is much larger than $[H_3O^+]$. Thus, we have neglected the concentration of H_3O^+ in these approximations.

We also assume that $[OH^-]$ is much smaller than c_{NH_4Cl} and c_{NH_3} so that

$$[NH_4^+] \approx c_{NH_4Cl} = 0.300 \text{ M}$$

$$[NH_3] \approx c_{NH_3} = 0.200 \text{ M}$$

When we substitute these expressions into the acid dissociation constant for NH_4^+, we have a relationship similar to Equation 9-29. That is,

$$[H_3O^+] = \frac{K_a \times [NH_4^+]}{[NH_3]} \approx \frac{5.70 \times 10^{-10} \times c_{NH_4Cl}}{c_{NH_3}}$$

$$= \frac{5.70 \times 10^{-10} \times 0.300}{0.200} = 8.55 \times 10^{-10} \text{ M}$$

To check the validity of our approximations, we calculate $[OH^-]$. Thus,

$$[OH^-] = \frac{1.00 \times 10^{-14}}{8.55 \times 10^{-10}} = 1.17 \times 10^{-5} \text{ M}$$

which is far smaller than c_{NH_4Cl} or c_{NH_3}. Finally, we write

$$pH = -\log(8.55 \times 10^{-10}) = 9.07$$

Sample Problems:

What is the ratio of sodium oxalate ($Na_2C_2O_4$) and sodium hydrogen oxalate ($NaHC_2O_4$) required for a buffer with a pH of 4.5? K_a of acid is equal to 6.4×10^{-5}

pH = pKa + log([A⁻]/[HA])

$$4.5 = 4.19 + \log(Na_2C_2O_4/NaHC_2O_4)$$

$$\log(Na_2C_2O_4/NaHC_2O_4) = 0.31$$

$$Na_2C_2O_4/NaHC_2O_4 = 10^{0.31} = \mathbf{2.04}$$

A buffer solution is made up of 0.600L of 0.12 M HClO, hypochlorous acid and 0.300L of 0.53 M sodium hypochlorite (NaClO). The K_a of HClO is 3.5×10^{-8}.

pH = pKa + log([A⁻]/[HA])

$$pH = 7.46 + \log(0.159/0.072)$$

pH = 7.80

For this solution, calculate the following:

a. Initial pH of the buffer solution.
b. Final pH of the solution if 0.030 moles NaOH is added to the buffer. (Assume addition of NaOH has no effect on volume.

pH = pKa + log([A⁻]/[HA])

$$pH = 7.46 + \log(0.189/0.042)$$

pH = 8.11

Importance and applications of [H+], pH and Buffers

- Water has slight propensity to dissociate into hydroxide ions and protons
    - Acidity, proton concentration of aqueous solutions reported through logarithmic pH scale

The **amount of H+** in a water-based solution tells us **how acidic the solution is.** Instead of writing the very small concentration of H+ directly, we use the **pH scale** to express it more conveniently.

The important relationship is:

More H+ → lower pH → more acidic

Less H+→ higher pH → more basic

The scale is called **logarithmic** because a change of 1 pH unit represents a **10-fold change in H+ concentration**.

- Buffers (esp. Bicarbonate) maintain pH of extracellular fluid (7.35-7.45)

A **buffer** is a system that helps **prevent large changes in pH**. In the human body, the **bicarbonate buffer system** is especially important. It helps keep the pH of the **extracellular fluid** within a very narrow range:

pH = 7.35 - 7.45

If too much acid is added, the buffer helps **absorb some of the H+**. If conditions become too basic, the buffer can help provide H+. This keeps the body's pH relatively stable.

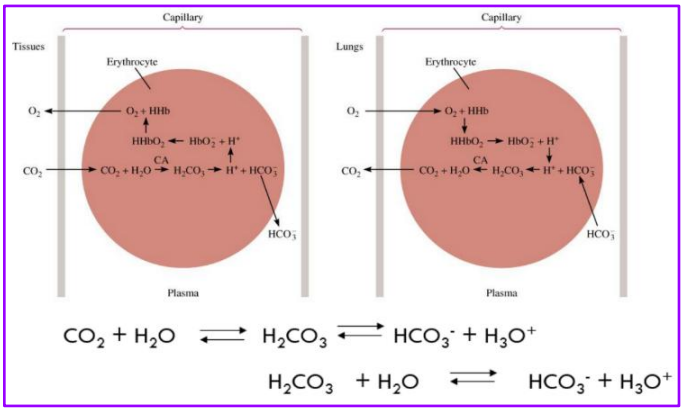
- Disturbance of acid-base balance in blood verified by measuring pH of arterial blood and the CO2 content of venous blood
    - Respiratory acidosis - restricted CO2 exhalation such as asthma, obesity or breathing problems due to trauma

Less CO_2 exhaled → CO_2 builds up → acid increases → pH↓

- Alkalosis (pH>7.45) follow vomiting of acidic gastric contents

Vomiting → Loss of stomach acid →H + ↓ → pH↑

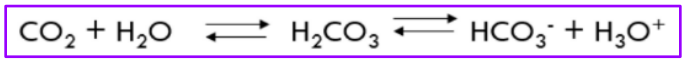
Acid-base balance in blood



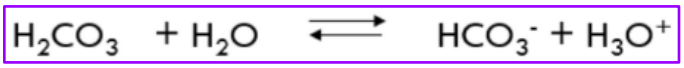
The picture above shows how the **bicarbonate buffer**

system carries CO_2 from the tissues to the lungs.

In the **tissues**, cells produce CO_2, which enters the red blood cells and combines with water to form carbonic acid.



This equation shows how **CO_2 is converted into bicarbonate in the blood and how this process helps control pH**. CO_2 combines with water to form carbonic acid. The carbonic acid (H2CO3) can then break apart into bicarbonate and hydronium. The arrows going in both directions mean that these reactions can **go forward or backward**, depending on what the body needs.



This shows **carbonic acid reacting with water**. Carbonic acid gives a hydrogen ion to water, forming **bicarbonate** and **hydronium**. The double arrow means the reaction can go **forward and backward**, which allows the bicarbonate buffer system to help **keep blood pH stable**.

----------END OF CHAPTER----------

MOCK QUIZ
ACID-BASE EQUILIBRIA

REVIEW SMART, GUYS! MWA

No.	Question	Choices	Correct answer	Rationalization
1	What is the pH of 2.5×10^{-3} M HCl?	A. 2.00 B. 2.60 C. 3.40 D. 11.40	**B. 2.60**	HCl is a strong acid, so pH = −log(2.5×10^{-3}) = 2.60. A implies a higher H^+ concentration; C implies a lower H^+ concentration; D is basic.
2	If pOH = 3.25 at 25°C, what is pH?	A. 3.25 B. 7.00 C. 10.75 D. 14.00	**C. 10.75**	pH = 14.00 − 3.25 = 10.75. A repeats pOH; B is neutral; D is not the calculated pH.
3	What is $[OH^-]$ when pH = 11.30?	A. 5.0×10^{-12} M B. 2.0×10^{-3} M C. 5.0×10^{-3} M D. 2.0×10^{-11} M	**B. 2.0×10^{-3} M**	pOH = 2.70, so $[OH^-] = 10^{-2.7} \approx 2.0 \times 10^{-3}$ M. A is approximately $[H^+]$; C is too high; D is too low.
4	What is the pH of 4.0×10^{-4} M NaOH?	A. 3.40 B. 4.00 C. 10.60 D. 11.40	**C. 10.60**	pOH = −log(4.0×10^{-4}) = 3.40; pH = 14 − 3.40 = 10.60. A is pOH; B and D do not match the calculation.
5	A solution has pH 5.00. What is $[H^+]$?	A. 1.0×10^{-5} M B. 1.0×10^{-9} M C. 1.0×10^{5} M D. 1.0×10^{-14} M	**A. 1.0×10^{-5} M**	$[H^+] = 10^{-5}$ M. B is approximately $[OH^-]$; C is physically unreasonable; D is Kw, not $[H^+]$.
6	What is the pH of 0.0010 M HNO_3?	A. 1.00 B. 2.00 C. 3.00 D. 11.00	**C. 3.00**	HNO_3 is a strong monoprotic acid, so $[H^+]$ = 0.0010 M and pH = 3.00. A and B imply higher concentrations; D is basic.
7	What is the pH of 0.010 M $Ba(OH)_2$, assuming complete dissociation?	A. 1.70 B. 2.00 C. 12.30 D. 12.00	**C. 12.30**	Each $Ba(OH)_2$ produces two OH^-, so $[OH^-]$ = 0.020 M. pOH ≈ 1.70 and pH ≈ 12.30. A is pOH; B ignores the second OH^-; D uses 0.010 M OH^-.

8	If pH changes from 4 to 2, $[H^+]$ increases by what factor?	A. 2 B. 10 C. 100 D. 1,000	**C. 100**	A two-unit decrease means a 100-fold increase in $[H^+]$. A treats pH linearly; B is one pH unit; D is three pH units.
9	At 25°C, which solution is acidic?	A. pH 8.2 B. pH 7.0 C. pH 6.8 D. pOH 3.0	**C. pH 6.8**	An acidic solution has pH below 7 at 25°C. A is basic; B is neutral; D gives pH 11 and is basic.
10	What is the approximate pH of 1.0×10^{-6} M HCl when water autoionization is considered?	A. Exactly 6.00 B. Slightly below 6.00 C. Exactly 7.00 D. Above 7.00	**B. Slightly below 6.00**	Accounting for water gives pH ≈ 5.996, slightly below 6. A neglects water; C ignores added acid; D incorrectly predicts a basic solution.
11	A 0.10 M weak acid has $K_a = 1.0 \times 10^{-5}$. Approximate its pH.	A. 1.00 B. 2.00 C. 3.00 D. 5.00	**C. 3.00**	$[H^+] \approx \sqrt{(K_aC)} = \sqrt{(10^{-6})} = 10^{-3}$ M, so pH = 3. A assumes complete ionization; B gives too much H^+; D gives too little.
12	A 0.20 M weak acid has $K_a = 4.0 \times 10^{-6}$. Approximate percent ionization.	A. 0.14% B. 0.45% C. 1.4% D. 4.0%	**B. 0.45%**	$[H^+] \approx \sqrt{(8.0 \times 10^{-7})} = 8.94 \times 10^{-4}$ M. Percent ionization ≈ (0.000894/0.20) × 100 = 0.45%. A underestimates; C and D overestimate.
13	Which acid is strongest?	A. $K_a = 10^{-8}$ B. $K_a = 10^{-5}$ C. $K_a = 10^{-3}$ D. $K_a = 10^{-10}$	**C. $K_a = 10^{-3}$**	A larger Ka means greater ionization and stronger acid. A, B, and D have smaller Ka values.
14	A 0.10 M weak base has $K_b = 1.0 \times 10^{-4}$. Approximate pH.	A. 3.00 B. 5.00 C. 9.00 D. 11.50	**D. 11.50**	$[OH^-] \approx \sqrt{(K_bC)} = 3.16 \times 10^{-3}$ M; pOH ≈ 2.50, so pH ≈ 11.50. A and B are acidic; C is too low.
15	If $K_a = 1.0 \times 10^{-6}$, what is the conjugate base's Kb at 25°C?	A. 1.0×10^{-20} B. 1.0×10^{-8}; C. 1.0×10^{-6} D. 1.0×10^{-14}	**B. 1.0×10^{-8}**	$K_b = K_w/K_a = 10^{-14}/10^{-6} = 10^{-8}$. A is too small; C incorrectly equates Ka and Kb; D is Kw.
16	What is the pKa of an acid with $K_a = 1.0 \times 10^{-4}$?	A. 2 B. 4 C. 10	**B. 4**	pKa = −log Ka = 4. A corresponds to Ka =

		D. 14		10^{-2}; C to 10^{-10}; D is pKw at 25°C.
17	Which acid is strongest based on pKa?	A. −2 B. 2 C. 5 D. 8	**A. −2**	Lower pKa means larger Ka and stronger acid. B and C are weaker; D is the weakest listed.
18	Which expression correctly represents Ka for HA in water?	A. $[H_3O^+][A^-]/[HA]$ B. $[HA]/([H_3O^+][A^-])$ C. $[H_3O^+][A^-]/([HA][H_2O])$ D. $[HA][H_2O]/([H_3O^+][A^-])$	**A. $[H_3O^+][A^-]/[HA]$**	Pure liquid water is omitted from the equilibrium expression. B and D reverse the expression; C incorrectly includes liquid water.
19	What is the conjugate base of $H_2PO_4^-$?	A. H_3PO_4 B. HPO_4^{2-} C. PO_4^{3-} D. OH^-	**B. HPO_4^{2-}**	Removing one H^+ gives HPO_4^{2-}. A is the conjugate acid; C requires losing more than one proton; D is unrelated.
20	In $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$, which is the Brønsted–Lowry acid?	A. NH_3 B. H_2O C. NH_4^+; D. OH^-	**B. H_2O**	Water donates H^+ to NH_3. NH_3 accepts H^+; NH_4^+ is the conjugate acid; OH^- is the conjugate base of water.
21	Which salt produces a basic aqueous solution?	A. NH_4Cl B. NaCl C. CH_3COONa D. $AlCl_3$	**C. CH_3COONa**	Acetate is the conjugate base of a weak acid and generates OH^- by hydrolysis. NH_4Cl and $AlCl_3$ are acidic; NaCl is approximately neutral.
22	What is the approximate pH of 0.10 M sodium acetate if Ka for acetic acid is 1.0×10^{-5}?	A. 5.00 B. 7.00 C. 8.00 D. 9.00	**D. 9.00**	$K_b = 10^{-14}/10^{-5} = 10^{-9}$. $[OH^-] \approx \sqrt{(10^{-10})} = 10^{-5}$ M; pOH = 5 and pH = 9. A is acidic; B ignores hydrolysis; C is too low.
23	Which combination forms an effective buffer?	A. HCl/NaCl B. NaOH/NaCl C. CH_3COOH/CH_3COONa D. HNO_3/KNO_3	**C. CH_3COOH/CH_3COONa**	A weak acid and its conjugate base form a buffer. The other pairs contain strong acids or bases and do not provide an

				effective conjugate pair.
24	A buffer has equal HA and A^- concentrations. What is its pH?	A. 7 B. pKa C. pKb D. 14 − pKa	**B. pKa**	Henderson–Hasselbalch gives pH = pKa + log(1) = pKa. A is not necessarily true; C and D are not the general buffer relation.
25	A buffer has pKa = 4.76, $[A^-]$ = 0.10 M, and [HA] = 0.20 M. Find pH.	A. 4.46 B. 4.76 C. 5.06 D. 9.24	**A. 4.46**	pH = 4.76 + log(0.10/0.20) ≈ 4.46. B assumes equal concentrations; C reverses the ratio; D is unrelated.
26	What happens when a small amount of HCl is added to an HA/A^- buffer?	A. A^- consumes H^+ B. HA consumes H^+ C. HCl produces OH^- D. Ka decreases	**A. A^- consumes H^+**	A^- reacts with H^+ to form HA. HA does not consume added H^+ in the buffer reaction; HCl supplies H^+; Ka is constant at fixed temperature.
27	What happens to acetic acid ionization when sodium acetate is added?	A. Increases B. Decreases C. Ka increases D. Solution must become strongly basic	**B. Decreases**	Added acetate is a common ion and shifts equilibrium toward HA. A is the opposite; Ka does not change at fixed temperature; D overstates the effect.
28	A buffer has pKa = 6.35 and $[A^-]/[HA]$ = 10. What is its pH?	A. 5.35 B. 6.35 C. 7.35 D. 10.00	**C. 7.35**	pH = 6.35 + log 10 = 7.35. A uses the reciprocal ratio; B assumes equal concentrations; D does not follow from the data.
29	Which buffer has the greatest capacity if the acid/base ratio is identical?	A. 0.001 M each B. 0.010 M each C. 0.100 M each D. All identical	**C. 0.100 M each**	Higher total buffer concentration gives greater capacity. A and B contain fewer buffer components; D confuses similar pH with equal capacity.

30	A buffer has pKa = 4.80. Strong acid is added, lowering $[A^-]/[HA]$. What happens to pH?	A. Increases B. Decreases C. Never changes D. Becomes 14	**B. Decreases**	A smaller ratio lowers the logarithmic term in Henderson–Hasselbalch. A is the opposite; C ignores concentration changes; D is unsupported.
31	What is the pH at equivalence in a strong acid–strong base titration at 25°C?	A. 0 B. 7 C. Above 7 D. Below 7	**B. 7**	The resulting salt does not appreciably hydrolyze, so pH is approximately 7. A is acidic; C and D describe other titration types.
32	At equivalence in a weak acid–strong base titration, pH is generally…	A. Below 7 B. Exactly 7 C. Above 7 D. Always 0	**C. Above 7**	The conjugate base hydrolyzes to form OH^-. A is typical of weak base–strong acid; B ignores hydrolysis; D is incorrect.
33	At the half-equivalence point of a weak acid titrated with strong base, what is true?	A. pH = pKa B. pH = 7 C. pH = pKb D. pH = 14	**A. pH = pKa**	Half the acid is converted to conjugate base, so $[HA] = [A^-]$ and pH = pKa. B is not universal; C uses the wrong constant; D is unrelated.
34	How much 0.100 M NaOH is needed to neutralize 25.0 mL of 0.100 M HCl?	A. 12.5 mL B. 25.0 mL C. 50.0 mL D. 100 mL	**B. 25.0 mL**	The reaction is 1:1 and both solutions have equal molarity, so equal volumes are required. A is half; C is double; D is four times the required volume.
35	Which indicator is often suitable for a weak acid–strong base titration?	A. Methyl orange B. Phenolphthalein C. Indicator changing at pH 1 D. Indicator changing at pH 3	**B. Phenolphthalein**	Phenolphthalein changes around pH 8.2–10, suitable for many basic equivalence points. The other indicators change too far into the acidic range.

36	What is the difference between endpoint and equivalence point?	A. They always match exactly B. Equivalence is stoichiometric completion; endpoint is observed indicator change C. Endpoint is initial pH D. Equivalence depends on color	**B. Equivalence is stoichiometric completion; endpoint is observed indicator change**	Equivalence is determined by reaction stoichiometry; endpoint is the observed indicator signal. A ignores indicator error; C is incorrect; D confuses observation with stoichiometry.
37	A 20.0 mL sample of 0.150 M monoprotic acid is neutralized by 0.100 M NaOH. What volume is needed?	A. 15.0 mL B. 20.0 mL C. 30.0 mL D. 45.0 mL	**C. 30.0 mL**	Moles acid = 0.0200 × 0.150 = 0.00300 mol. Volume base = 0.00300/0.100 = 0.0300 L = 30.0 mL. A and B are too small; D is too large.
38	During titration of a weak acid with strong base, what dominates at equivalence?	A. Only HA B. Mostly A^- from the salt C. Excess HA and no base D. Only OH^- from titrant	**B. Mostly A^- from the salt**	The weak acid has been converted to its conjugate base. A ignores neutralization; C describes before equivalence; D requires excess strong base.
39	Which condition makes the weak-acid approximation $C - x \approx C$ least reliable?	A. Very small Ka, high concentration B. Very small Ka, moderate concentration C. Relatively large Ka, very low concentration D. Small Ka, high concentration	**C. Relatively large Ka, very low concentration**	A larger Ka and low concentration can produce substantial fractional ionization. The other conditions generally make ionization a smaller fraction of initial concentration.
40	A buffer initially contains 0.20 mol HA and 0.20 mol A^-. Add 0.05 mol OH^-. What are the final moles?	A. HA 0.25, A^- 0.15 B. HA 0.15, A^- 0.25 C. HA 0.20, A^- 0.25 D. HA 0.15, A^- 0.20	**B. HA 0.15 mol; A^- 0.25 mol**	OH^- converts HA into A^-. HA decreases to 0.15 mol and A^- increases to 0.25 mol. A reverses the changes; C and D omit one change.
41	If $Ka = 1.0 \times 10^{-4}$, what is Kb of the conjugate base at 25°C?	A. 1.0×10^{-10} B. 1.0×10^{-14} C. 1.0×10^{-8} D. 1.0×10^{-18}	**A. 1.0×10^{-10}**	$Kb = Kw/Ka = 10^{-14}/10^{-4} = 10^{-10}$. B is Kw; C is too large; D is too small.

42	Which statement about Ka is correct?	A. Changes with concentration B. Changes with common-ion addition C. Changes with temperature D. Equals initial acid concentration	**C. Changes with temperature**	Ka is temperature-dependent. Concentration and common-ion changes shift equilibrium composition, not Ka at constant temperature. D confuses a constant with concentration.
43	Why does a weak acid generally have a higher pH than a strong acid at the same initial concentration?	A. It has a larger Ka B. It ionizes partially and produces less H^+ C. It contains no hydrogen D. Strong acids produce no ions	**B. It ionizes partially and produces less H^+**	Weak acids ionize incompletely and typically produce less H^+. A reverses the strength relationship; C is false; D is opposite to strong-acid behavior.
44	What is the main error in calculating the pH of 0.10 M weak acid as −log(0.10)?	A. pOH must always be calculated first B. Assumes complete ionization C. Uses 14 − pKa D. Must multiply by 100	**B. Assumes complete ionization**	The calculation assumes $[H^+] = 0.10$ M, not generally true for a weak acid. A is unnecessary; C is not a general pH equation; D is incorrect.
45	A buffer ratio $[A^-]/[HA]$ changes from 1 to 100. By how much does pH change?	A. Decreases by 2 B. Increases by 1 C. Increases by 2 D. Increases by 100	**C. Increases by 2**	The log term changes from log 1 = 0 to log 100 = 2. A has the wrong direction; B underestimates; D treats a logarithmic change as linear.
46	Which is a conjugate acid–base pair?	A. HCl and NaOH B. NH_4^+ and NH_3 C. H_2O and NaCl D. H_2SO_4 and OH^-	**B. NH_4^+ and NH_3**	NH_4^+ and NH_3 differ by one proton. The other pairs do not differ by exactly one H^+.
47	Which statement describes a neutral solution at 25°C?	A. $[H^+] = 0$ B. $[OH^-] = 0$ C. $[H^+] = [OH^-] = 10^{-7}$ M D. pH must be 0	**C. $[H^+] = [OH^-] = 10^{-7}$ M**	At neutrality, $[H^+] = [OH^-]$, each 10^{-7} M at 25°C. A and B are incorrect; D is strongly acidic.
48	A weak base has Kb = 1.0×10^{-5}. What is pKb?	A. 2 B. 5 C. 9 D. 14	**B. 5**	pKb = −log Kb = 5. A corresponds to 10^{-2}; C to 10^{-9}; D is pKw, not this base's pKb.

49	Which change generally increases buffer capacity?	A. Diluting both components equally B. Increasing both component concentrations while maintaining their ratio C. Removing conjugate base D. Making one component nearly absent	**B. Increasing both concentrations while maintaining their ratio**	Greater total concentrations increase the amount of acid/base neutralized. A lowers capacity; C and D weaken the buffer.
50	What best describes the common-ion effect?	A. Changes Ka at constant temperature B. Shifts equilibrium when an ion already present is added C. Always makes pH 7 D. Prevents equilibrium	**B. Shifts equilibrium when an ion already present is added**	A is false at fixed temperature; C is not universal; D misunderstands equilibrium. A common ion changes the equilibrium position.

CHEMISTRY LAB I UNIT 4

MEME WALL



ACID BASE TITRATION

THEORY

Titration

- is a process or method to determine the concentration of a substance in an unknown solution, in which a known reagent is added to an unknown solution to produce a known reaction such as change in color or electrical measurement.

ADVANTAGES OF TITRATION OVER GRAVIMETRIC ANALYSIS

❑ Superior Speed

❑ Convenient with a little sacrifice in accuracy and Precision

❑Less expensive and less materials to use

Requirements for a successful titration

- **Stoichiometric** → definite, reproducible relationship between reacting substances.
- **Quantitative** → reaction proceeds to completion when an amount of titrant equivalent to the analyte is added.
- **Fast** → slow reactions make the equivalence point difficult to determine.

USES OF ACID-BASE TITRATIONS

- **Pregnancy testing** – used to measure the amount of hCG in blood.
- **Aquarium water testing** – used to monitor pH, ammonia, nitrites, nitrates, and phosphates in aquarium water.
- **Blood sugar testing** – used in measuring blood glucose.
- **Pharmacy & medicine** – pharmacists use titration to achieve the desired mix/proportion of compounds; doctors use it to determine the proper proportion of medicines in IV drips.
- **Food industry** – used to test:
    - Free and unsaturated fatty acids
    - Trace amounts of water
    - Salt and sugar content
    - Vitamin C and E concentration
    - Readiness of products such as wine and cheese
    - Estimated chain length of fatty acids in fats
- **Biology laboratories** – used to determine the appropriate concentration of chemicals/anaesthetic agents for test animals.

ACID-BASE TITRATIONS: CHEMICAL REACTIONS

Neutralization Reaction

- A neutralization reaction occurs when acids and bases react to form a neutral solution.
- OH^- ions neutralize excess H^+ ions to form water (H_2O).
- In acid–base titration, the acid and base react according to a definite stoichiometric relationship.

General idea:

$$H^+ + OH^- \rightarrow H_2O$$

Important Terms to Remember:

Titrant

- Solution with a known concentration
- Added from the burette

Analyte

- Sample with an unknown concentration
- The solution being titrated

Equivalence Point

- Point where the acid and base have exactly neutralized one another
- The amounts of H^+ and OH^- are stoichiometrically equivalent

Indicator

- Substance added to the receiving vessel that produces a color change when the titration reaches the endpoint

End Point

- Point at which the indicator changes color

Key Relationship

At the equivalence point:

Amount of acid = amount of base (according to the reaction stoichiometry)

For a 1:1 acid–base reaction:

$$C_aV_a = C_\beta V_\beta$$

TYPES OF ACID-BASE TITRATION

(A) Strong Acid-Strong Base Titration

Example: HCl versus NaOH

(B) Strong Acid-Weak Base Titration

Example: HCl versus ammonia

(C) Weak Acid-Strong Base Titration

Example: NaOH versus acetic acid

- **Acidimetry**: When sample is an acid
- **Alkalimetry:** When sample is a base

ACID-BASE TITRATIONS

- In **titration**, a solution of accurately known concentration, called a **standard solution**, is added gradually to another solution of unknown concentration, until the chemical reaction between the two solutions is complete.
- *Sodium hydroxide* is one of the bases commonly used in the laboratory. We can standardize the sodium hydroxide solution by titrating it against an acid solution of accurately known concentration.
- The acid often chosen for this task is a monoprotic acid called *potassium hydrogen phthalate* (KHP), for which the molecular formula is KHC8H4O4 (molar mass: 204.2 g).

$$KHC_8H_4O_4(aq) + NaOH(aq) \longrightarrow KNaC_8H_4O_4(aq) + H_2O(l)$$

and the net ionic equation is

$$HC_8H_4O_4^-(aq) + OH^-(aq) \longrightarrow C_8H_4O_4^{2-}(aq) + H_2O(l)$$

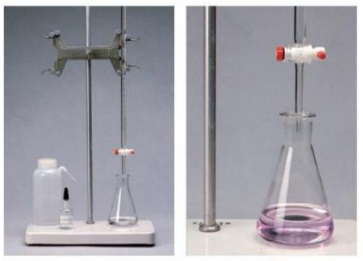


Figure 4.21
(a) Apparatus for acid-base titration. A NaOH solution is added from the buret to a KHP solution in an Erlenmeyer flask.
(b) A reddish-pink color appears when the equivalence point is reached. The color here has been intensified for visual display.

The procedure for the titration is shown in Figure 4.21.

- First, a known amount of KHP is transferred to an Erlenmeyer flask and some distilled water is added to make up a solution.
- Next, NaOH solution is carefully added to the KHP solution from a buret until we reach the equivalence point, that is, the point at which the acid has completely reacted with or been neutralized by the base.
- The equivalence point is usually signaled by a sharp change in the color of an indicator in the acid solution.
- In acid-base titrations, indicators are substances that have distinctly different colors in acidic and basic media.

- One commonly used indicator is phenolphthalein, which is colorless in acidic and neutral solutions but reddish pink in basic solutions.
- At the equivalence point, all the KHP present has been neutralized by the added NaOH and the solution is still colorless. However, if we add just one more drop of NaOH solution from the buret, the solution will immediately turn pink because the solution is now basic.

SAMPLE PROBLEM: In a titration experiment, a student finds that 23.48 mL of a NaOH solution are needed to neutralize 0.5468 g of KHP. What is the concentration (in molarity) of the NaOH solution?

Solution First we calculate the number of moles of KHP consumed in the titration:

$$\text{moles of KHP} = 0.5468 \text{ g KHP} \times \frac{1 \text{ mol KHP}}{204.2 \text{ g KHP}}$$
$$= 2.678 \times 10^{-3} \text{ mol KHP}$$

Because 1 mol KHP ≏ 1 mol NaOH, there must be 2.678×10^{-3} mole of NaOH in 23.48 mL of NaOH solution. Finally, we calculate the number of moles of NaOH in 1 L of the solution or the molarity as follows:

$$\text{molarity of NaOH soln} = \frac{2.678 \times 10^{-3} \text{ mol NaOH}}{23.48 \text{ mL soln}} \times \frac{1000 \text{ mL soln}}{1 \text{ L soln}}$$
$$= 0.1141 \text{ mol NaOH/1 L soln} = 0.1141 \, M$$

SAMPLE PROBLEM: The sodium hydroxide solution standardized in Example 4.11 is used to titrate 25.00 mL of a sulfuric acid solution. The titration requires 43.79 mL of the 0.1172 M NaOH solution to completely neutralize the acid. What is the concentration of the H_2SO_4 solution?

$$2NaOH(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$$

Solution First, we calculate the number of moles of NaOH contained in 43.79 mL of solution:

$$43.79 \text{ mL} \times \frac{1 \text{ L soln}}{1000 \text{ mL soln}} \times \frac{0.1172 \text{ mol NaOH}}{\text{L soln}} = 5.132 \times 10^{-3} \text{ mol NaOH}$$

From the stoichiometry we see that 1 mol H_2SO_4 ≏ 2 mol NaOH. Therefore, the number of moles of H_2SO_4 reacted must be

$$5.132 \times 10^{-3} \text{ mol NaOH} \times \frac{1 \text{ mol } H_2SO_4}{2 \text{ mol NaOH}} = 2.566 \times 10^{-3} \text{ mol } H_2SO_4$$

From the definition of molarity [see Equation (4.1)], we have

$$\text{molarity} = \frac{\text{moles of solute}}{\text{liters of soln}}$$

So the molarity of the H_2SO_4 solution is

$$\frac{2.566 \times 10^{-3} \text{ mol } H_2SO_4}{25 \text{ mL} \times (1 \text{ L}/1000 \text{ mL})} = 0.1026 \, M \, H_2SO_4$$

EXERCISE: If 60.2 mL of 0.427 M KOH solution are required to neutralize 10.1 mL of H_2SO_4 solution, what is the concentration of the H_2SO_4 solution in molarity?

1.27 M

----------END OF CHAPTER----------

i

Question 1

In a standardization experiment, a student uses 28.50 mL of an $NaOH$ solution to completely neutralize 0.6134 g of KHP ($KHC_8H_4O_4$, molar mass = 204.2 g/mol). What is the molar concentration of the $NaOH$ solution?

Question 2

A volume of 35.00 mL of a hydrochloric acid (HCl) solution is titrated with $0.1250M\ NaOH$. If 29.15 mL of the $NaOH$ solution are required to reach the equivalence point, what is the concentration of the HCl solution?

Question 3

If 45.8 mL of $0.215M\ Ba(OH)_2$ solution are required to neutralize 25.0 mL of a phosphoric acid (H_3PO_4) solution, what is the concentration of the H_3PO_4 solution?

SOLUTIONS:

Question 1

In a standardization experiment, a student uses 28.50 mL of an NaOH solution to completely neutralize 0.6134 g of KHP (KHC8H4O4, molar mass = 204.2 g/mol). What is the molar concentration of the NaOH solution?

Solution:

1. Calculate the moles of KHP: Moles of KHP = 0.6134 g KHP x (1 mol KHP / 204.2 g KHP) = 3.0039 x 10^-3 mol KHP
2. Use the stoichiometric mole ratio (1 mol KHP : 1 mol NaOH) to find the moles of NaOH: Moles of NaOH = 3.0039 x 10^-3 mol
3. Calculate the molarity of the NaOH solution: Molarity = (3.0039 x 10^-3 mol NaOH) / 0.02850 L = 0.1054 M

Question 2

A volume of 35.00 mL of a hydrochloric acid (HCl) solution is titrated with 0.1250 M NaOH. If 29.15 mL of the NaOH solution are required to reach the equivalence point, what is the concentration of the HCl solution? *(Balanced equation: NaOH(aq) + HCl(aq) -> NaCl(aq) + H2O(l))*

Solution:

1. Calculate the moles of NaOH used: Moles of NaOH = 0.02915 L x 0.1250 mol/L = 3.6438 x 10^-3 mol NaOH

2. Use stoichiometry (1 mol NaOH : 1 mol HCl) to find the moles of HCl: Moles of HCl = 3.6438 x 10^-3 mol
3. Calculate the molarity of the HCl solution: Molarity = (3.6438 x 10^-3 mol) / 0.03500 L = 0.1041 M

Question 3

If 45.8 mL of 0.215 M Ba(OH)2 solution are required to neutralize 25.0 mL of a phosphoric acid (H3PO4) solution, what is the concentration of the H3PO4 solution? *(Balanced equation: 3 Ba(OH)2(aq) + 2 H3PO4(aq) -> Ba3(PO4)2(s) + 6 H2O(l))*

Solution:

1. Calculate the moles of Ba(OH)2 used: Moles of Ba(OH)2 = 0.0458 L x 0.215 mol/L = 9.847 x 10^-3 mol Ba(OH)2
2. Use stoichiometry (3 mol Ba(OH)2 : 2 mol H3PO4) to find moles of H3PO4: Moles of H3PO4 = 9.847 x 10^-3 mol Ba(OH)2 x (2 mol H3PO4 / 3 mol Ba(OH)2) = 6.5647 x 10^-3 mol
3. Calculate the molarity of the H3PO4 solution: Molarity = (6.5647 x 10^-3 mol) / 0.0250 L = 0.263 M

CHEMISTRY LAB I UNIT 8

MEME WALL



PROPERTIES OF GASES

- Gases have relatively large spaces between particles.
- They have very low density compared to liquids or solids.
- Gases are highly compressible compared to liquids and solids.
- Gases can expand or contract to fill their container.

Some Common Compounds That Are Gasses at Room Temperature

Formula	Name	Characteristics
HCN	Hydrogen cyanide	Very toxic, slight odor of bitter almonds
H_2S	Hydrogen sulfide	Very toxic, odor of rotten eggs
CO	Carbon monoxide	Toxic, colorless, odorless
CO_2	Carbon dioxide	Colorless, odorless
CH_4	Methane	Colorless, odorless, flammable
C_2H_4	Ethene (Ethylene)	Colorless, ripens fruit
C_3H_8	Propane	Colorless, odorless, bottled gas
N_2O	Nitrous oxide	Colorless, sweet odor, laughing gas
NO_2	Nitrogen dioxide	Toxic, red-brown, irritating odor
NH_3	Ammonia	Colorless, pungent odor
SO_2	Sulfur dioxide	Colorless, irritating odor

UNITS OF PRESSURE

A generalization that describes in mathematical terms the relationships among the amount of **gas (mole)**, **pressure**, **temperature (in Kelvin)**, and **volume of gas (in Liters)**.

PRESSURE

- The force applied per unit area on an object.
- The average pressure at sea level is called standard pressure and is equal to any of the following:
- **1 atm = 760 mm Hg = 760 torr = 101.3245 kPa = 1.01325 bar**
- **1 atm = 14.7 psi**

PRESSURE OF GAS

The force that creates pressure is that which is exerted by the gas molecules or atoms as they constantly collide with the walls of their container

- **Barometer:** A device used to measure atmospheric pressure

Example:

1. A geochemist heats a limestone (CaCO3) sample and collects the CO2 released in an evacuated flask attached to a closed-end manometer. After the system comes to room temperature, Dh = 291.4 mm Hg. Calculate the CO2 pressure in torrs, atmospheres, and kilopascals.

SOLUTION:

$$291.4 \text{ mm Hg} \times \frac{1 \text{ torr}}{1 \text{ mm Hg}} = 291.4 \text{ torr}$$

$$291.4 \text{ torr} \times \frac{1 \text{ atm}}{760 \text{ torr}} = 0.3834 \text{ atm}$$

$$0.3834 \text{ atm} \times \frac{101.325 \text{ kPa}}{1 \text{ atm}} = 38.85 \text{ kPa}$$

THE PRESSURE-VOLUME RELATIONSHIP: BOYLE'S LAW

- The pressure of a fixed amount of gas maintained at constant temperature is inversely proportional to the volume of the gas.
- When two measurements are inversely proportional, one gets smaller as the other gets larger. Boyle's law can be expressed mathematically as:
  P1 × V1 = P2 × V2

Example:

1. A volume of air occupying 12.0 dm3 at 98.9 kPa is compressed to a pressure of 119.0 kPa. The temperature remains constant. What is the new volume?

Problem Strategy Putting the data for the problem in tabular form, you see what data you have and what you must find. This will suggest the method of solution.

$V_i = 12.0\ \text{dm}^3$ $P_i = 98.9\ \text{kPa}$
$V_f = \ ?$ $P_f = 119.0\ \text{kPa}$
} Temperature and moles remain constant.

Because P and V vary but temperature and moles are constant, you use Boyle's law.

Solution Application of Boyle's law gives

$$V_f = V_i \times \frac{P_i}{P_f} = 12.0\ \text{dm}^3 \times \frac{98.9\ \cancel{\text{kPa}}}{119.0\ \cancel{\text{kPa}}} = \mathbf{9.97\ dm^3}$$

THE TEMPERATURE-VOLUME RELATIONSHIP: CHARLE'S LAW

- The volume occupied by any sample of gas at a constant pressure is directly proportional to the absolute temperature.
- Volume of a fixed amount of gas is directly related to its temperature (in Kelvin = °C +273.15) Constant P and n
- **FORMULA:**

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Example:

1. Earlier we found that the total volume of oxygen that can be obtained from a particular tank at 1.00 atm and 21.0°C is 785 L (including the volume remaining in the tank). What would be this volume of oxygen if the temperature had been 28.0 °C?

Solution $T_i = (21 + 273)\text{K} = 294\ \text{K}$
$T_f = (28 + 273)\text{K} = 301\ \text{K}$

Following is the data table:

$V_i = 785\ \text{L}$ $P_i = 1.00\ \text{atm}$ $T_i = 294\ \text{K}$
$V_f = ?$ $P_f = 1.00\ \text{atm}$ $T_f = 301\ \text{K}$

Note that T varies and P remains constant, so V must change. These are the conditions needed to apply Charles's law.

$$V_f = V_i \times \frac{T_f}{T_i} = 785\ \text{L} \times \frac{301\ \cancel{\text{K}}}{294\ \cancel{\text{K}}} = \mathbf{804\ L}$$

THE TEMPERATURE-PRESSURE RELATIONSHIP: GAY-LUSSAC'S LAW

- The pressure of a given mass of gas varies directly with the absolute temperature of the gas, when the volume is kept constant.
- The pressure and absolute temperature (K) of a gas are directly related at constant mass & volume
- **FORMULA:**

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

Example:

1. A gas' pressure is 765 torr at 23°C. At what temperature will the pressure be 560 torr?

GIVEN:
P_1 = 765 torr
T_1 = 23°C = 296K
P_2 = 560. torr
T_2 = ?

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

(765 torr)T_2 = (560. torr)(296K)

T_2 = 217 K = -56°C

THE VOLUME-MOLE RELATIONSHIP: AVOGADRO'S LAW

- Avogadro's Law states that volume is directly proportional to moles when temperature and pressure are unchanging
- **FORMULA:**

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

Example:

1. How many moles of gas were added to a balloon to cause it to expand from 5.0 L to 35.2 L at constant temperature and pressure? Initially there were 0.30 moles in the balloon.

GIVEN:
V_1 = 5.0 L
V_2 = 35.2 L
n_1 = 0.30 moles

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

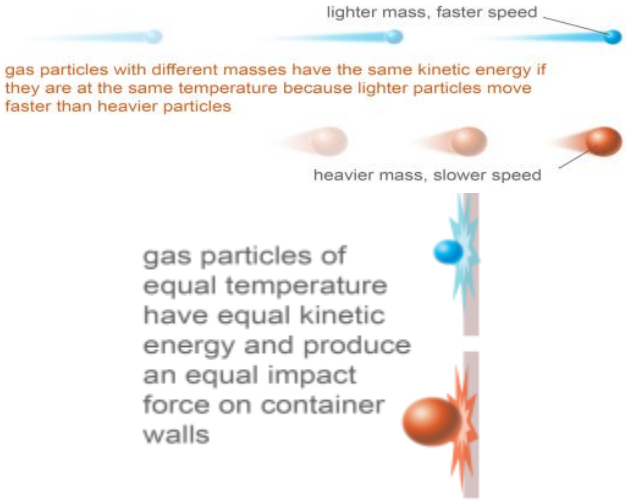
$$\frac{5.0\ \text{L}}{0.30\ \text{moles}} = \frac{35.2\ \text{L}}{n_2}$$

$$n_2 = \frac{35.2\ \cancel{\text{L}} \times 0.30\ \text{moles}}{5.0\ \cancel{\text{L}}} = 2.112 \xrightarrow{\text{round}} 2.1\ \text{moles}$$

ANSWER: n added = n2−n1 = 2.112mol − 0.30 mol = **1.812mol**

MOLAR VOLUME

- If two gases are at the same temperature, they must have particles with the same average kinetic energy.
- The two different gas particles will exert the same force when they hit the wall of their container.
- This is why at standard temperature and pressure **(STP, 0 °C (273.15 K) and 1.00 atm)** one mole of any gas occupies 22.4 L. This is known as the **molar volume** of a gas.



RELATING VOLUME, TEMPERATURE, AND PRESSURE: COMBINED GAS LAW

- **Boyle's law ($V \propto 1/P$)** and **Charles's law ($V \propto T$)** can be combined and expressed in a single statement:
  - The volume occupied by a given amount of gas is proportional to the absolute temperature divided by the pressure **($V \propto T/P$)**.
- **FORMULA:**

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Example:

1. Using the Dumas method, 39.8 mg of caffeine gives 10.1 cm3 of nitrogen gas at 23°C and 746 mmHg. What is the volume of nitrogen at 0°C and 760 mmHg?

$$V_i = 10.1\ \text{cm}^3 \qquad P_i = 746\ \text{mmHg} \qquad T_i = 296\ \text{K}$$
$$V_f = \quad ? \qquad P_f = 760\ \text{mmHg} \qquad T_f = 273\ \text{K}$$

$$V_f = V_i \times \frac{P_i}{P_f} \times \frac{T_f}{T_i} = 10.1\ \text{cm}^3 \times \frac{746\ \cancel{\text{mmHg}}}{760\ \cancel{\text{mmHg}}} \times \frac{273\ \cancel{\text{K}}}{296\ \cancel{\text{K}}} = \mathbf{9.14\ cm^3}$$

KINETIC MOLECULAR THEORY OF GASES

1. Gases consist of large numbers of molecules that are in continuous, random motion.
2. The combined volume of all the molecules of the gas is negligible relative to the total volume in which the gas is contained.
3. Attractive and repulsive forces between gas molecules are negligible.
4. Energy can be transferred between molecules during collisions but, as long as temperature remains constant, the average kinetic energy of the molecules does not change with time.
5. The average kinetic energy of the molecules is proportional to the absolute temperature. At any given temperature the molecules of all gases have the same average kinetic energy.

GAS

- Physical state characterized by a complete dominance of kinetic energy **(disruptive forces)** over potential energy **(cohesive forces)**
- Attractive forces among particles are very weak and are considered to be zero
- Particles move essentially independently of one another in a random manner

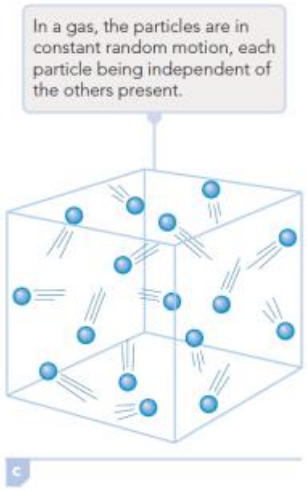


Figure 7.3(c) - Distance Between Particles and Particle Movement in a Gas

THE IDEAL GAS LAW

- The equation **PV=nRT**, which combines all of the gas laws, is called the **ideal gas law**.
- UNIVERSAL GAS CONSTANT **(R=0.0821 L.atm/mol.K or R=8.315 dm3.kPa/mol.K)**
- Describes the relationships among the four variables – temperature, pressure, volume, and molar amount for a gaseous substance

$$\text{Boyle's law: } V \propto \frac{1}{P} \quad (\text{at constant } n \text{ and } T)$$
$$\text{Charles's law: } V \propto T \quad (\text{at constant } n \text{ and } P)$$
$$\text{Avogadro's law: } V \propto n \quad (\text{at constant } P \text{ and } T)$$

Example:

1. Sulfur hexafluoride (SF6) is a colorless, odorless, very unreactive gas. Calculate the pressure (in atm) exerted by 1.82 moles of the gas in a steel vessel of volume 5.43 L at 45.0 °C.

$PV = nRT$

$$P = \frac{nRT}{V}$$
$$= \frac{(1.82\ \text{mol})(0.0821\ \text{L} \cdot \text{atm/K} \cdot \text{mol})(45 + 273)\ \text{K}}{5.43\ \text{L}}$$
$$= 8.75\ \text{atm}$$

DALTON'S LAW OF PARTIAL PRESSURES

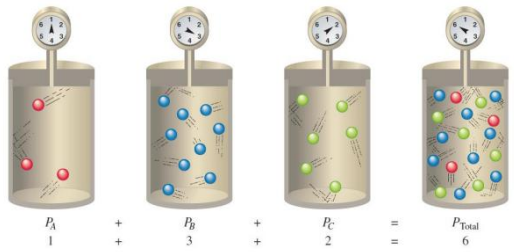
- The total pressure of a mixture of gases is just the sum of the pressures that each gas would exert if it were present alone.
- For a mixture of gases in a container, **PTotal = PA + PB + PC + ...**
- The total pressure exerted by a mixture of gases is the sum of the partial pressures of the individual gases present
- The individual partial pressures follow the ideal gas law. For component A,

$P_AV=n_ART$

n_A *is the number of moles of component A*

$$P_t = (n_1 + n_2 + n_3 + \cdots)\left(\frac{RT}{V}\right) = n_t\left(\frac{RT}{V}\right)$$

Figure 7.14 - Illustration of Dalton's Law of Partial Pressures



Example:

1. A 1.00-L sample of dry air at 25.0°C and 786 mmHg contains 0.925 g N2, plus other gases including oxygen, argon, and carbon dioxide. What is the partial pressure (in mmHg) of N2 in the air sample?

Solution a. Each gas in a mixture follows the ideal gas law. To calculate the partial pressure of N_2, you convert 0.925 g N_2 to moles N_2.

$$0.925\ \text{g}\ \cancel{\text{N}_2} \times \frac{1\ \text{mol}\ \text{N}_2}{28.0\ \text{g}\ \cancel{\text{N}_2}} = 0.0330\ \text{mol}\ \text{N}_2$$

You substitute into the ideal gas law (noting that 25°C is 298 K).

$$P_{N_2} = \frac{n_{N_2}RT}{V}$$

$$= \frac{0.0330\ \cancel{\text{mol}} \times 0.0821\ \cancel{\text{L}}\cdot\text{atm}/(\cancel{\text{K}}\cdot\cancel{\text{mol}}) \times 298\ \cancel{\text{K}}}{1.00\ \cancel{\text{L}}}$$

= 0.808 atm = 614 mm Hg

----------END OF CHAPTER----------

1. Which statement best defines pressure?
A. The amount of force exerted by an object regardless of its surface area
B. The force applied per unit area on an object
C. The amount of energy contained in a substance
D. The rate at which particles collide with one another

2. Which of the following represents standard atmospheric pressure correctly?
A. $1atm = 760mmHg = 101.325kPa$
B. $1atm = 760mmHg = 14.7kPa$
C. $1atm = 760torr = 1.01325kPa$
D. $1atm = 101.325bar = 14.7psi$

3. What is the primary cause of gas pressure inside a sealed container?
A. The gravitational attraction between gas molecules
B. The expansion of gas molecules as they gain volume
C. The collisions of gas particles with the container walls
D. The attractive forces between particles and the container

4. Which instrument is used to measure atmospheric pressure?
A. Manometer
B. Barometer
C. Thermometer
D. Calorimeter

5. Which property distinguishes gases from most liquids and solids?
A. Gas particles have no kinetic energy.
B. Gas particles are arranged in fixed positions.
C. Gases have relatively high density and low compressibility.
D. Gases have large spaces between particles and are highly compressible.

6. A gas occupies $12.0L$ at $98.9kPa$. If it is compressed to $119.0kPa$ at constant temperature, what is its new volume?
A. $10.0L$
B. $14.4L$
C. $8.31L$
D. $11.0L$

7. Which condition must remain constant for Boyle's law to apply?
A. Pressure and temperature
B. Volume and amount of gas
C. Temperature and amount of gas
D. Pressure and amount of gas

8. A gas sample is compressed to half its original volume while its temperature and amount remain constant. What happens to its pressure?
A. It decreases to one-fourth its original value.
B. It doubles.
C. It remains unchanged.
D. It decreases to half its original value.

9. A gas has a volume of $4.0L$ at $2.0atm$. If its volume increases to $10.0L$ at constant temperature, what is its final pressure?
A. $5.0atm$
B. $0.20atm$
C. $0.80atm$
D. $1.25atm$

10. Which statement correctly describes Charles's law?
A. Volume is inversely proportional to pressure at constant temperature.
B. Pressure is directly proportional to temperature at constant volume.
C. Volume is directly proportional to absolute temperature at constant pressure.
D. Volume is directly proportional to the number of moles at constant temperature and pressure.

11. A gas occupies $785L$ at 21.0°C. What volume will it occupy at 28.0°C, assuming constant pressure and amount of gas?
A. $767L$
B. $804L$
C. $1{,}047L$
D. $748L$

12. Why must temperatures be expressed in Kelvin when applying Charles's law?
A. Kelvin eliminates the need to account for pressure.
B. Kelvin makes gas volume independent of temperature.
C. Celsius temperatures cannot be used in any gas-law calculation, even after conversion.
D. Gas-law proportionalities involving temperature require an absolute temperature scale.

13. A gas occupies $3.0L$ at $300K$. If the temperature increases to $450K$ at constant pressure, what is the new volume?
A. $2.0L$
B. $4.5L$
C. $6.0L$
D. $3.5L$

14. Which statement describes Gay-Lussac's law?
A. Pressure is directly proportional to absolute temperature at constant volume.
B. Pressure is inversely proportional to absolute temperature at constant volume.
C. Volume is directly proportional to pressure at constant temperature.
D. Pressure is directly proportional to volume at constant temperature.

15. A gas has a pressure of $765torr$ at 23°C. At what temperature, approximately, will its pressure be $560torr$ if volume and amount remain constant?
A. $218K$
B. $296K$
C. $23K$
D. $404K$

16. A sealed rigid container is heated. Assuming the amount of gas remains unchanged, which change is expected?
A. Both pressure and volume decrease.
B. Pressure decreases while volume remains constant.
C. Pressure increases while volume remains constant.
D. Volume increases while pressure remains constant.

17. According to Avogadro's law, which relationship holds when temperature and pressure are constant?
A. Volume is inversely proportional to the number of moles.
B. Volume is directly proportional to the number of moles.
C. Pressure is directly proportional to the number of moles, regardless of volume.
D. Temperature is inversely proportional to the number of moles.

18. A balloon expands from $5.0L$ to $35.2L$ at constant temperature and pressure. If it initially contains $0.30mol$ of gas, how many moles were added?
A. $0.30mol$
B. $1.81mol$
C. $2.11mol$
D. $2.41mol$

19. A gas sample contains $2.0mol$ and occupies $8.0L$. How much volume will $5.0mol$ occupy at the same temperature and pressure?
A. $3.2L$
B. $8.0L$
C. $13.0L$
D. $20.0L$

20. Which combination of gas laws forms the basis of the combined gas law?
A. Boyle's, Charles's, and Gay-Lussac's laws B. Avogadro's law and Dalton's law only C. Boyle's law and Avogadro's law only D. Charles's law and Dalton's law only
21. Which equation correctly represents the combined gas law for a fixed amount of gas?

A. $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

B. $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$, with temperature in Celsius

C. $\frac{P_1T_1}{V_1} = \frac{P_2T_2}{V_2}$

D. $P_1V_1T_1 = P_2V_2T_2$

22. A gas occupies $10.1cm^3$ at 23°C and $746mmHg$. What is its approximate volume at 0°C and $760mmHg$?
A. $10.8cm^3$
B. $9.21cm^3$
C. $9.48cm^3$
D. $10.4cm^3$

23. A gas initially has a volume of $6.0L$, pressure of $1.5atm$, and temperature of $300K$. It changes to $3.0L$ and $400K$. What is its final pressure?
A. $0.75atm$
B. $1.0atm$
C. $2.0atm$
D. $6.0atm$

24. At standard temperature and pressure (STP), what volume is occupied by one mole of an ideal gas according to the provided notes?
A. $1.00L$
B. $22.4L$
C. $24.0L$
D. $273.15L$

25. Which conditions define STP in the provided material?
A. 25°C and $1.00atm$
B. 0°C and $760kPa$
C. $273.15K$ and $1.00atm$
D. $273.15K$ and $1.00bar$

26. Two different gases are at the same temperature. According to the kinetic molecular theory, which statement is correct?
A. Their molecules have the same average kinetic energy.
B. Their molecules have the same average speed regardless of molar mass.
C. Their molecules exert identical gravitational forces on the container.
D. Their molecules must have the same molar mass.

27. Which statement is NOT one of the assumptions of the kinetic molecular theory of gases presented in the notes?
A. Gas particles undergo continuous, random motion.
B. The combined volume of gas particles is negligible compared with the container volume.
C. Attractive and repulsive forces between gas particles are negligible.
D. Gas particles remain stationary between collisions.

28. According to the kinetic molecular theory, what happens to the average kinetic energy of gas molecules when the absolute temperature increases?
A. It decreases inversely with temperature.
B. It increases in direct proportion to absolute temperature.
C. It remains constant as long as the pressure is unchanged.
D. It becomes independent of the type of gas and also independent of temperature.

29. Which statement best describes the gas phase in the provided notes?
A. Potential energy completely dominates kinetic energy.
B. Strong attractive forces hold particles in fixed positions.
C. Kinetic energy dominates potential energy, allowing particles to move relatively independently.
D. Particles vibrate around fixed positions and cannot move randomly.

30. What is the ideal gas law?
A. $PV = nRT$
B. $P_1V_1 = P_2V_2$
C. $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
D. $P_{\text{total}} = P_A + P_B$

31. Which value is an appropriate universal gas constant when pressure is in atmospheres and volume is in liters?
A. $8.315L \cdot atmmol^{-1}K^{-1}$
B. $0.0821L \cdot atmmol^{-1}K^{-1}$
C. $760L \cdot atmmol^{-1}K^{-1}$

D. $273.15L \cdot atmmol^{-1}K^{-1}$

32. A sample of $1.82mol$ of sulfur hexafluoride occupies a $5.43L$ vessel at 45°C. What pressure does it exert, approximately?
A. $4.37atm$
B. $5.43atm$
C. $8.75atm$
D. $12.4atm$

33. A gas occupies $4.00L$ at $2.00atm$ and $300K$. How many moles of gas are present? Use $R = 0.0821L \cdot atmmol^{-1}K^{-1}$.
A. $0.163mol$
B. $0.325mol$
C. $0.488mol$
D. $1.22mol$

34. A mixture contains three gases with partial pressures of $120mmHg$, $250mmHg$, and $180mmHg$. What is the total pressure?
A. $130mmHg$
B. $250mmHg$
C. $370mmHg$
D. $550mmHg$

35. Which statement best expresses Dalton's law of partial pressures?
A. The total pressure of a gas mixture is the average of the individual gas pressures.
B. The total pressure is the sum of the partial pressures of the gases present.
C. Each gas contributes pressure in direct proportion to its volume alone.
D. The pressure of each gas equals the total pressure divided by the number of gases.

36. A $1.00L$ sample of dry air at 25°C and $786mmHg$ contains $0.925g$ of nitrogen gas. What is the approximate partial pressure of nitrogen? Use a molar mass of $28.02g/mol$.
A. $307mmHg$
B. $786mmHg$
C. $614mmHg$
D. $925mmHg$

37. A mixture of oxygen and nitrogen exerts a total pressure of $900mmHg$. If oxygen contributes $240mmHg$, what is the partial pressure of nitrogen, assuming these are the only two gases present?
A. $240mmHg$
B. $660mmHg$
C. $900mmHg$
D. $1{,}140mmHg$

38. A gas sample has a pressure of $1.00atm$. What is this pressure in kilopascals using the standard pressure conversion provided in the notes?
A. $76.0kPa$
B. $100.0kPa$
C. $101.325kPa$
D. $760kPa$

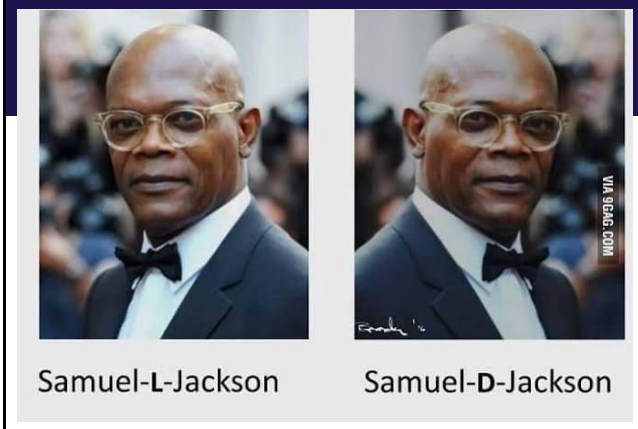
39. A geochemist collects carbon dioxide in an evacuated flask attached to a closed-end manometer. If the height difference is $291.4mmHg$, what is the carbon dioxide pressure in atmospheres, approximately?
A. $0.383atm$
B. $0.765atm$
C. $1.00atm$
D. $2.61atm$

40. A rigid container holds a fixed amount of ideal gas. Its absolute temperature doubles while its volume remains constant. Which statement correctly describes the resulting pressure and explains why?
A. The pressure is halved because the particles spread farther apart.
B. The pressure remains constant because the amount of gas is unchanged.
C. The pressure doubles because pressure is directly proportional to absolute temperature at constant volume.
D. The pressure quadruples because both molecular speed and collision frequency double independently.

No.	Answer	No.	Answer	No.	Answer	No.	Answer
1	B	11	B	21	A	31	B
2	A	12	D	22	C	32	C
3	C	13	B	23	D	33	B
4	B	14	A	24	B	34	D
5	D	15	A	25	C	35	B
6	A	16	C	26	A	36	C
7	C	17	B	27	D	37	B
8	B	18	B	28	B	38	C
9	C	19	D	29	C	39	A
10	C	20	A	30	A	40	C

CHEMISTRY I UNIT 11

MEME WALL

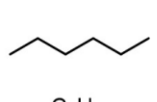


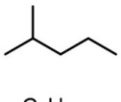
MAIN TOPIC

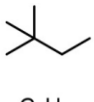
- Isomers - same mol formula, diff structure
  - Have different phys and chem properties
- Constitutional - different bonding/connectivity
- Stereo - diff 3d arrangement
  - Geometric - restricted rotation

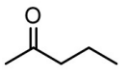
CONSTITUTIONAL ISOMERS

- Different connections
- More carbon atoms = more isomers can be formed from the same molecular formula

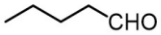

C_6H_{14}


C_6H_{14}

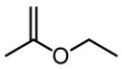

C_6H_{14}



$C_5H_{10}O$



$C_5H_{10}O$



$C_5H_{10}O$

- Skeletal isomers (commonly referred to as chain isomers)
- Positional isomers (also known as regioisomers)
- Functional isomers (sometimes referred to as functional group isomers)

STEREOISOMERS

- Same connectivity, different 3D arrangement

Geometric isomers - restricted rotation caused by rigid double bond or ring formation

- Double bonds
  - Cis - same substituents on the same side of the double bond (Z)
  - Trans (opposite) - same substituents on the opposite side of the double bond (E)
- Bonds in a ring
  - Cis - same substituents on the same side of the ring (Z)
  - Trans (opposite) - same substituents on the opposite side of the ring (E)

CONFORMATIONAL ISOMER

- Difference due to rotation
- Conformation -

In ring formation

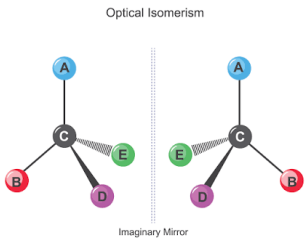
- Chair -
  - More stable; minimizes repulsion
  - Found in nature esp in carbohydrates
- Boat -

Not ring

- Staggered - spread out
  - More stable; minimizes repulsion
- Eclipsed - H will align

OPTICAL ISOMER

- Differ in 3D arrangement of atoms such that they are nonsuperimposable.
  - • One isomer rotates light clockwise (**dextrorotatory**, +)
  - • The other rotates light counter-clockwise (**levorotatory**, -).
  - • Optical isomers usually arise from the presence of a **chiral center**—a carbon atom bonded to four different groups.



CHIRALITY

- **Chiral objects** are objects with left-handed and right-handed forms
- **Achiral objects** - objects that have superimposable mirror images.
- **Nonsuperimposable mirror images** - a mirror image that is not

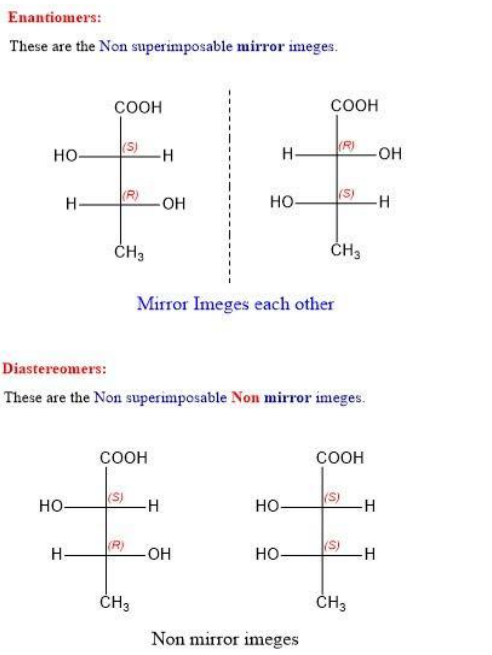
the same as the image itself - chiral objects have nonsuperimposable mirror images

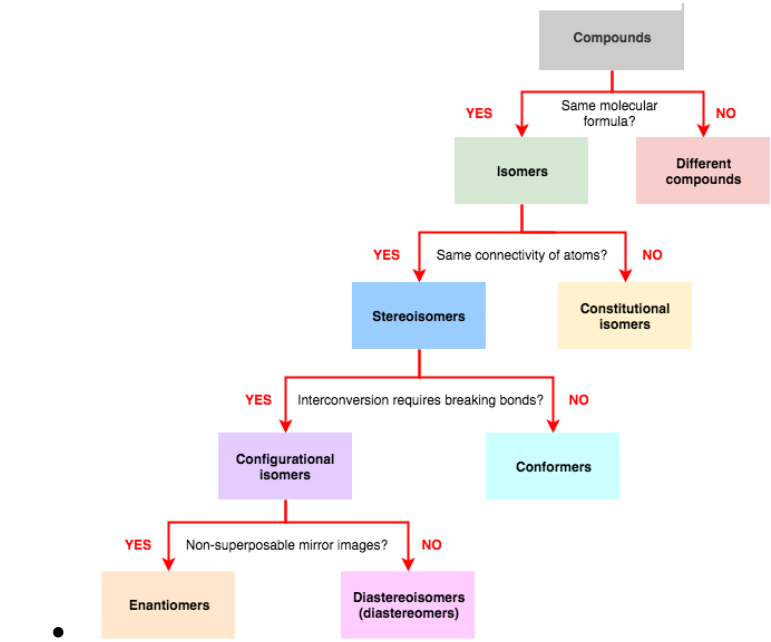
- **Chiral Molecules -** Molecules with one chiral center would have 2 enantiomers
- Chiral molecules would have **2^n** enantiomers (where n is the number of chiral centers)

ENANTIOMERS AND DIASTEREOSOMERS

- **Stereoisomers** are either enantiomers or diastereomers.

- **Enantiomers**: are a pair of optical isomers that are **non-superimposable mirror images** of each other. They have:
    1. The same molecular formula
    2. The same connectivity
    3. The same physical properties (melting point, boiling point, density)
    4. Difference: They rotate plane-polarized light in opposite directions (one dextro +, other levo -)
- **Diastereosomers:** are chiral molecules but are not mirror images of each other and are **not superimposable**. They have noticeably different physical and chemical properties, such as boiling point, solubility, and reactivity.
- Common when **chiral centers** are more than two





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----------END OF CHAPTER----------

UNIT 11: ISOMERS - Mock Quiz

PART I - FUNDAMENTALS OF ISOMERISM

Question	Choices
1. What are isomers?	A. Compounds with different molecular formulas but identical structures B. Compounds with the same molecular formula but different structures C. Compounds with the same physical and chemical properties in all cases D. Compounds containing the same number of carbon atoms only
2. Why can isomers have different physical and chemical properties?	A. They always contain different elements B. They always have different molecular formulas C. Their atoms may be connected or arranged differently D. They always have different numbers of electrons
3. Which type of isomer has the same molecular formula but a different connectivity of atoms?	A. Constitutional isomer B. Geometric isomer C. Conformational isomer D. Optical isomer
4. Which type of isomer has the same molecular formula and connectivity but a different three-dimensional arrangement?	A. Skeletal isomer B. Positional isomer C. Functional group isomer D. Stereoisomer
5. Which of the following is NOT a type of stereoisomerism discussed in the presentation?	A. Geometric isomerism B. Conformational isomerism C. Optical isomerism D. Positional isomerism
6. Which statement about constitutional isomers is correct?	A. They must be mirror images B. They have different atom connectivity C. They differ only by rotation around a single bond D. They must contain a chiral center

7. According to the presentation, what generally happens to the number of possible constitutional isomers as the number of carbon atoms increases?	A. The number always becomes one B. The number decreases C. More isomers can be formed D. The molecular formula must change to an aromatic formula
8. Which type of isomerism results from restricted rotation, usually around a double bond or within a ring?	A. Geometric B. Conformational C. Skeletal D. Functional group

PART II - CONSTITUTIONAL ISOMERS

Question	Choices
9. Two compounds have the same molecular formula and functional group, but their carbon skeletons differ. What type of constitutional isomers are they?	A. Positional B. Skeletal C. Functional group D. Geometric
10. Two compounds have the same carbon skeleton and functional group, but the functional group is located at different positions. What type of isomers are they?	A. Skeletal B. Optical C. Positional D. Conformational
11. Two compounds have the same molecular formula but belong to different functional groups. What type of isomers are they?	A. Functional group B. Skeletal C. Geometric D. Conformational
12. Which compound is NOT an isomer of hexane, C_6H_{14}?	A. 2,2-Dimethylbutane B. 2,2-Dimethylpentane C. 3-Methylpentane D. 2,3-Dimethylbutane
13. Which pair best illustrates positional isomerism?	A. Butane and 2-methylpropane B. Propan-1-ol and propan-2-ol C. Ethanol and dimethyl ether D. Ethene and ethane

14. Which pair best illustrates functional group isomerism?	A. Butane and 2-methylpropane B. Propan-1-ol and propan-2-ol C. Ethanol and dimethyl ether D. But-1-ene and but-2-ene

PART III - GEOMETRIC ISOMERISM

Question	Choices
15. In a molecule with a double bond, what does cis mean when the relevant substituents are the same?	A. They are on opposite sides of the double bond B. They are on the same side of the double bond C. They rotate freely around the double bond D. They form non-superimposable mirror images
16. In a molecule with a double bond, what does trans mean when the relevant substituents are the same?	A. They are on the same side of the double bond B. They are attached to the same carbon C. They are on opposite sides of the double bond D. They must be part of a ring
17. In a cyclic compound, what does cis indicate about two substituents?	A. They are on the same side of the ring B. They are on opposite sides of the ring C. They are attached to the same carbon D. They rotate freely around a double bond
18. In a cyclic compound, what does trans indicate about two substituents?	A. They are on the same side of the ring B. They are on opposite sides of the ring C. They must be identical D. They must be mirror images
19. Why can 1,1-dichloroethene not exhibit cis-trans isomerism?	A. It has no carbon atoms B. It contains a single bond only C. One double-bonded carbon has two identical substituents D. It contains two different elements
20. Which pair of compounds is used in the presentation as an example of geometric isomers?	A. Ethanol and dimethyl ether B. Cis-1,2-dichloroethene and trans-1,2-dichloroethene C. Butane and 2-methylpropane D. Staggered and eclipsed cyclohexane

21. A double-bonded compound has two identical substituents attached to one of its double-bonded carbon atoms. Which conclusion is correct?	A. It must be trans B. It must be cis C. It cannot have any stereoisomers of any kind D. It cannot exhibit cis-trans isomerism at that double bond
22. Two identical substituents attached to a double bond are drawn on opposite sides of that bond. Which description applies?	A. Cis B. Trans C. Skeletal D. Conformational

PART IV - CONFORMATIONAL ISOMERISM

Question	Choices
23. Conformational isomers interconvert through what process?	A. Breaking and reforming the molecular formula B. Rotation around single bonds C. Changing the number of protons D. Changing the connectivity of atoms permanently
24. Which two conformations of cyclohexane are illustrated in the presentation?	A. Cis and trans B. R and S C. Chair and boat D. Staggered and cis
25. Which two conformations of ethane are illustrated in the presentation?	A. Chair and boat B. Cis and trans C. Enantiomer and diastereomer D. Staggered and eclipsed
26. Which statement best distinguishes conformational isomers from geometric isomers?	A. Conformational isomers can interconvert by rotation around single bonds B. Conformational isomers always have different molecular formulas C. Geometric isomers always have different atom connectivity D. Geometric isomers are always mirror images

PART V - OPTICAL ISOMERISM AND CHIRALITY

Question	Choices
27. What is a chiral object?	A. An object whose mirror image is always superimposable B. An object whose mirror image is non-superimposable C. An object that contains only carbon atoms D. An object that cannot form mirror images
28. What is an achiral object?	A. An object with four different groups attached to carbon B. An object whose mirror image is non-superimposable C. An object whose mirror image is superimposable on itself D. An object that always rotates plane-polarized light
29. What is a chiral center typically described as in the presentation?	A. A carbon atom bonded to four different groups B. A carbon atom bonded to four identical groups C. Any carbon atom in a double bond D. A carbon atom bonded to two hydrogen atoms only
30. Optical isomers are described as which of the following?	A. Compounds with different molecular formulas B. Compounds with different atom connectivity C. Non-superimposable mirror images of each other D. Compounds that differ only by rotation around single bonds
31. What does dextrorotatory (+) indicate?	A. Rotation of plane-polarized light counterclockwise B. Rotation of plane-polarized light clockwise C. No rotation of plane-polarized light D. The absence of a chiral center
32. What does levorotatory (−) indicate?	A. Rotation of plane-polarized light clockwise B. No rotation of plane-polarized light C. Rotation of plane-polarized light counterclockwise D. A change in molecular formula
33. According to the presentation, how many enantiomers are associated with a molecule having one chiral center?	A. One B. Two C. Four D. Eight

PART VI - ENANTIOMERS AND DIASTEREOMERS

Question	Choices
34. What are enantiomers?	A. Isomers with different molecular formulas B. Non-superimposable mirror images of each other C. Isomers that differ only in the position of a functional group D. Conformations that differ by rotation around single bonds
35. Which statement about enantiomers is consistent with the presentation?	A. They have different molecular formulas B. They have different atom connectivity C. They have the same molecular formula and connectivity but rotate plane-polarized light in opposite directions D. They are never mirror images
36. Which statement best describes diastereomers?	A. They are always superimposable mirror images B. They are non-superimposable stereoisomers that are not mirror images of each other C. They must have different molecular formulas D. They differ only in carbon skeleton connectivity
37. Compared with enantiomers, diastereomers generally have:	A. Identical physical and chemical properties in all cases B. Different molecular formulas C. Noticeably different physical and chemical properties D. No stereochemical differences
38. Two stereoisomers are non-superimposable mirror images. What is their relationship?	A. Diastereomers B. Enantiomers C. Positional isomers D. Skeletal isomers
39. Two stereoisomers are not mirror images of each other. What is their relationship, assuming they are stereoisomers?	A. Enantiomers B. Functional group isomers C. Diastereomers D. Constitutional isomers

40. Using the formula 2^n for the maximum number of stereoisomers, how many stereoisomers can a molecule with three chiral centers have?	A. Three B. Six C. Eight D. Nine

ANSWER KEY

Question	Answer	Question	Answer	Question	Answer	Question	Answer
1	B	11	A	21	D	31	B
2	C	12	B	22	B	32	C
3	A	13	B	23	B	33	B
4	D	14	C	24	C	34	B
5	D	15	B	25	D	35	C
6	B	16	C	26	A	36	B
7	C	17	A	27	B	37	C
8	A	18	B	28	C	38	B
9	B	19	C	29	A	39	C
10	C	20	B	30	C	40	C

CHEMISTRY LAB

MEME WALL



MELTING POINT

- The **melting point** of a substance is the temperature at which its solid form is in equilibrium with its liquid form.
- A pure crystalline organic substance usually has a sharp melting point that melts completely over a short temperature range. The melting point of pure substances is influenced by the **forces that hold each particle** (electrostatic forces and intermolecular forces) and also its **size.**
- The **stronger the forces that hold each particle, the higher the melting point.** Furthermore, bigger molecules tend to have higher melting point than small molecules due to surface area.

CAPILLARY METHOD





- The melting point range of the substance can be determined by introducing a finely powdered material in the capillary that is sealed at one end.
- Two temperatures are recorded:
    - the temperature at which substance begins to liquefy
    - the temperature at which the substance becomes completely liquefied.

FACTORS AFFECTING THE MELTING POINT OF COMPOUNDS

1. **The bond that holds the particles together.**

- Ionic bonds hold ions together while only intermolecular forces hold molecules together.
- In comparison, ionic bond is stronger than intermolecular forces (London dispersion, dipole-dipole, H-bond) therefore, ionic compounds have high melting point than covalent compounds

2. **Intermolecular forces.**

- Between common intermolecular forces, the H-bond has the strongest hold to molecules. Next to it is a dipole-dipole that holds polar molecules.
- And lastly, the London dispersion forces that hold non-polar molecules.
- Thus this also means, polar molecules have higher melting point than non-polar molecules.

3. **Size of molecule**

- Usually, we compute the molecular weight of a compound to predict its size.
- The larger the size of the molecule would mean high melting point given the same intermolecular forces.
- This is because of the wider surface area that each molecule has in contact with each other.

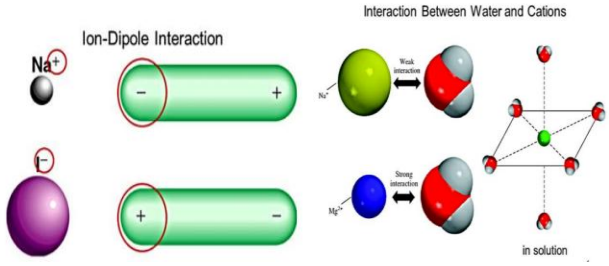
INTERMOLECULAR FORCES

- **Intramolecular Forces** - The forces between atoms or ions inside a compound such as covalent and ionic bonding (determines the stability of a compound)
    - $H\text{-}O\text{-}H \rightarrow 2H + O$ E = 920 kJ/mol
- **Intermolecular Forces** - The forces between individual particles. (determines the phase a compound is in at room temperature)
    - $H2O(l) \rightarrow H2O(g)$ E=40.7 kJ/mol

1. **Ion-Ion Intermolecular Forces**

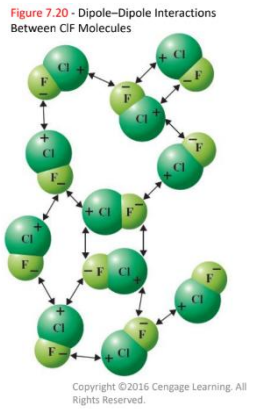
- ionic compounds have strong Intermolecular forces
- solids at room temperature and they have high m.p.
- ionic compounds with multiply charged ions generally have higher m.p. than those with singly charged ions

2. Ion-Dipole Intermolecular Forces



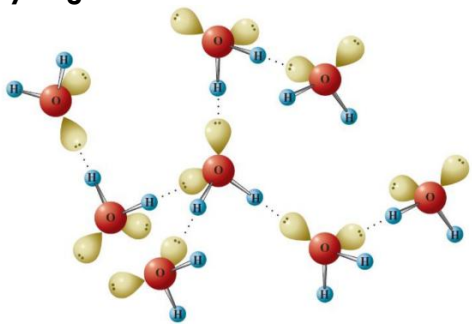
- Attractive forces between an **ion** and a **polar molecule**

3. Dipole–Dipole Interaction



- An intermolecular force that occurs between polar molecules
- Molecules with dipole moments can attract each other electrostatically by lining up so that the positive and negative ends are close to each other
- The greater the polarity of the molecules, the greater the strength of the dipole-dipole interactions

4. Hydrogen Bonds



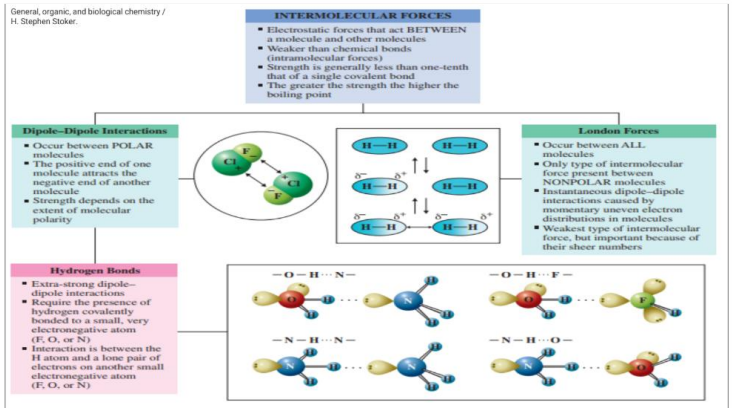
- Unusually strong dipole–dipole interactions are observed among hydrogen-containing molecules in which hydrogen is covalently bonded to a highly electronegative element of small atomic size (fluorine, oxygen, and nitrogen)
- **Factors Accounting for the Extra Strength of Hydrogen Bonds:**
  1. The highly electronegative element to which hydrogen is covalently bonded attracts the bonding electrons to such a degree that the hydrogen atom is left with a significant charge
  2. The small size of the "bare" hydrogen n ucleus allows it to approach closely, and be strongly attracted to, a lone pair of electrons on the electronegative atom of another molecule

5. Dispersion Forces



The more electrons a molecule/atom has, the stronger the dispersion force

- weak attractive forces that all molecules have.
- dispersion forces are the only intermolecular interaction that small,
- non-polar molecules undergo. (generally gases at room temp.)



PHYSICAL PROPERTIES

PHYSICAL PROPERTIES - properties of a substance that can be observed or measured without changing what the substance is chemically.

Important Physical Properties of Organic Compounds

1. melting point
2. boiling point

3. solubility

Intermolecular Forces (IMFs) — attractions between molecules. The strength of these attractions affects the melting point, boiling point, and solubility of a compound.

Relative Strength of Common Intermolecular Forces

H-bonding > dipole-dipole > London dispersion

MELTING POINT AND BOILING POINT

Melting Point (m.p.) — the temperature at which an organic substance changes from a solid into a liquid.

Boiling Point (b.p.) — the temperature at which an organic substance changes from a liquid into a gas (vapor).

Effect of Noncovalent Interactions — differences in the melting and boiling points of organic molecules show how the attractions between molecules affect their physical properties.

EFFECT OF CHAIN LENGTH

Increasing Chain Length — as the carbon chain becomes longer, the melting point and boiling point generally tend to increase within a homologous series.

Homologous Series — a group of organic compounds with the same general structure and similar chemical properties, where each successive member differs by a $-CH_2-$ unit.

General Trend in Unbranched Alkanes — as the number of carbon atoms increases, the boiling point generally increases, and the melting point also tends to increase.

Table 2.5: Physical Properties of Some Unbranched Alkanes

Name	Condensed Structural Formula	Melting Point (°C)	Boiling Point (°C)	Density of Liquid* (g/mL at 0°C)
Methane	CH_4	−182	−164	(a gas)
Ethane	CH_3CH_3	−183	−88	(a gas)
Propane	$CH_3CH_2CH_3$	−190	−42	(a gas)
Butane	$CH_3(CH_2)_2CH_3$	−138	0	(a gas)
Pentane	$CH_3(CH_2)_3CH_3$	−130	36	0.626
Hexane	$CH_3(CH_2)_4CH_3$	−95	69	0.659
Heptane	$CH_3(CH_2)_5CH_3$	−90	98	0.684
Octane	$CH_3(CH_2)_6CH_3$	−57	126	0.703
Nonane	$CH_3(CH_2)_7CH_3$	−51	151	0.718
Decane	$CH_3(CH_2)_8CH_3$	−30	174	0.730

*For comparison, the density of H_2O is 1 g/mL at 4°C

EFFECT OF BRANCHING

Branching — when the carbon chain has carbon groups attached to the main chain instead of forming one continuous chain.

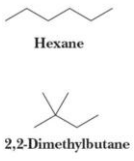
Effect of Branching on Boiling Point — increasing branching makes the molecule more spherical, which decreases its surface area.

Surface Area — the amount of molecular surface that can come into contact with other molecules.

Important: The PPT specifically explains the effect of branching in terms of boiling point. The melting-point values of the C_6H_{14} isomers do not follow a simple "more branching = lower melting point" rule.

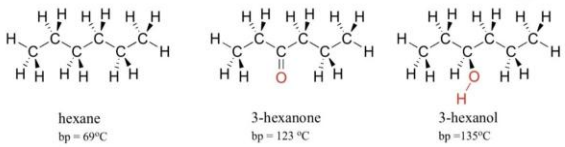
Table 2.6: Physical Properties of the Isomeric Alkanes of Molecular Formula C_6H_{14}

Name	Boiling Point (°C)	Melting Point (°C)	Density (g/mL)
Hexane	68.7	−95	0.659
2-Methylpentane	60.3	−154	0.653
3-Methylpentane	63.3	−118	0.664
2,3-Dimethylbutane	58.0	−129	0.661
2,2-Dimethylbutane	49.7	−98	0.649



❑ The strength of intermolecular hydrogen bonding and dipole- dipole interactions is reflected in higher boiling points.

❑ The hydrogen from one molecule is strongly attracted to a lone pair of electrons on the oxygen of another molecule.



❑O-H more polar than N-H, so alcohols have stronger hydrogen bonding

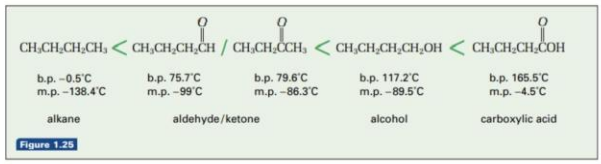
$CH_3CH_2CH_3$	$CH_3CH_2NH_2$	CH_3CH_2OH
Propane	Ethanamine	Ethanol
M.W. = 44 g/mol	M.W. = 45 g/mol	M.W. = 46 g/mol
b.p. = −42.2° C	b.p. = 16.6° C	b.p. = 78.5° C

❑Hydrogen bonding increases the attractive forces between molecules of ethanol, which results in a higher boiling point than dimethyl ether.

CH_3CH_2OH	CH_3OCH_3
Ethanol	**Dimethyl ether**
bp 78 ° C	bp −24 ° C

❑ Because of the strong hydrogen bonds between molecules, the melting and boiling points of carboxylic acids are very high.

Figure 1.25 compares the melting and boiling points of a carboxylic acid with the melting and boiling points of other organic compounds.



COMPARISON BETWEEN IONIC AND COVALENT SUBSTANCES

Ionic Compound — a compound whose particles are held together by ionic bonds.

Covalent Compound — a compound made of molecules whose particles are held together by intermolecular forces.

Ionic Bond — the strong attraction between oppositely charged ions.

NaCl (Sodium Chloride) — an ionic compound with a melting point of about 801 °C.

Salicylic Acid — a covalent organic compound with a melting point of about 158 °C.

Experimental Observation — in the PPT experiment, NaCl melted last, while salicylic acid melted first.

Why Ionic Compounds Generally Have Higher Melting Points — ionic bonds are stronger than the intermolecular forces holding molecular covalent substances together.

Experimental Results

Substance	Observations	Forces that hold particles
NaCl (Right)	Melts last	Ionic bond
Salicylic acid (Left)	Melts first	Covalent bond



Sodium Chloride

- Commonly known as **salt**
- Ionic compound
- Chemical formula: NaCl
- Molar mass: **58.44 g/mol**
- Melting point: **801°C**
- **Soluble in water**
- One of the most abundant minerals on Earth
- An essential nutrient for many animals and plants



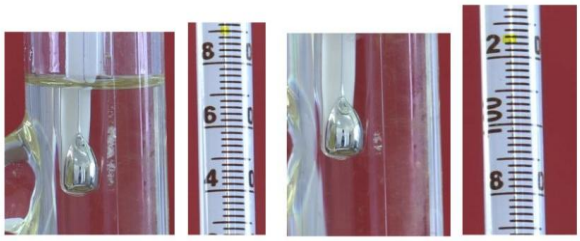




Substance	T1	T2	Intermolecular Forces that hold particles
Sodium Chloride	-	-	Ion-dipole forces

Salicylic Acid

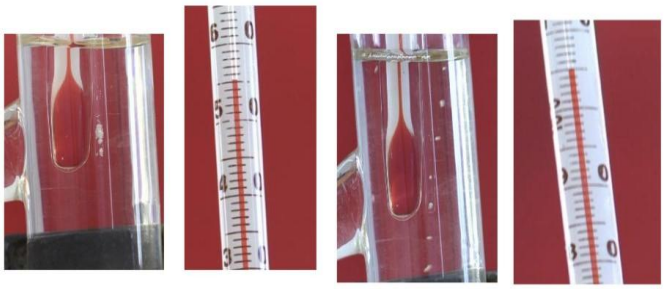
- Used as a component of aspirin
- Known as a fever and pain reliever
- Used in anti-inflammatory medicines
- IUPAC name: **2-Hydroxybenzoic acid**
- Chemical formula: **$C_7H_6O_3$**
- Molar mass: **138.121 g/mol**
- Melting point: approximately **158°C**



Substance	T1	T2	Intermolecular Forces that hold particles
Salicylic Acid	157	159	2 H-bonds (-OH and –COOH group)

Benzoic Acid

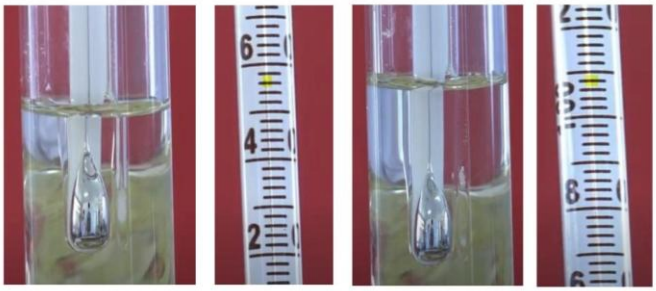
- A white, crystalline organic compound
- Belongs to the carboxylic acid family
- Also referred to as **benzene-carbonic-acid**
- Chemical formula: **$C_7H_6O_2$** or **C_6H_5COOH**
- **Very soluble in alcohol**
- Molecular weight: **122.12 g/mol**
- Melting point: approximately **123°C**



Substance	T1	T2	Intermolecular Forces that hold particles
Benzoic Acid	122	124	H-bond bigger molecules than oxalic

Oxalic Acid

- A **toxic organic acid**
- Also called **ethanedioic acid**
- Molecular formula: **$C_2H_2O_4$**
- Molar mass: **90.03 g/mol**
- Melting point: approximately **101°C**
- **Water soluble**



Substance	T1	T2	Intermolecular Forces that hold particles
Oxalic Acid	100	102	H-bond small molecule

Note:

Melting point (MP): Temperature at which solid and liquid phases exist in equilibrium.
Pure crystalline substances: Usually have a sharp, narrow melting-point range.
Impurities: Usually lower the melting point and broaden the melting range.
Stronger particle attractions: Generally result in higher melting points.
Ionic compounds: Generally have higher melting points than molecular covalent compounds due to strong electrostatic attractions.
Intermolecular forces: Hydrogen bonding > dipole–dipole > London dispersion forces (general comparison of common molecular IMFs).
Molecular size: Larger molecules generally have higher melting points when other structural factors are comparable.
Increasing carbon-chain length: Melting and boiling points generally increase within a homologous series.
Molecular structure and crystal packing: Isomers can have different melting points despite having the same molecular formula.
London dispersion forces: Present in all molecules; generally stronger with increasing electron count and polarizability.
Hydrogen bonding: Occurs when H is covalently bonded to N, O, or F and interacts with an electronegative atom on another molecule.
Carboxylic acids: Can form strong hydrogen-bonding interactions, contributing to their physical properties.

Experimental measurements

- T_1: Temperature when the sample begins to melt.
- T_2: Temperature when the sample has completely melted.
- Melting range: $\Delta T = T_2 - T_1$

Remember: Melting-point trends are affected by both intermolecular/electrostatic forces and molecular structure. Stronger attractions are a useful guide, but melting point is not determined by one factor alone.

----------END OF CHAPTER----------

Question: *May solvings po ba here sa Melting Point?*

Meron, pero **very minimal lang ang solving sa Melting Point PPT**, hahahaha 😭

Ang actual computation na kailangan mong alam ay:

$$\Delta T = T_2 - T_1$$

- T_1 = temperature when the sample *starts melting*
- T_2 = temperature when the sample is *completely melted*
- ΔT = melting range

The rest is mostly conceptual questions and comparing trends na—like how impurities affect melting point, how intermolecular forces affect it, and why different isomers can have different melting points.

READ THROUGH THESE NA LANG PO IF YOU HAVE THE TIME (Review well, goats) 🫶

QUESTION / SOLVING	ANSWER + RATIONALE
PART I — FUNDAMENTALS AND PROCEDURE	
1. What is melting point? A. Solid/liquid equilibrium temperature B. Liquid-to-gas temperature C. Decomposition temperature D. Dissolving temperature E. The answer is not in the choices	A — Melting point is where solid and liquid phases exist in equilibrium.
2. A pure crystalline organic substance usually has: A. Wide range B. Sharp, narrow range C. No complete melting D. MP always below 0°C E. The answer is not in the choices	B — Pure crystalline substances usually melt over a short range.
3. Which forces influence MP according to the PPT? A. Gravity/magnetism B. Nuclear/frictional C. Electrostatic/intermolecular D. Centrifugal/gravity E. The answer is not in the choices	C — Both electrostatic and intermolecular forces affect MP.
4. Stronger particle attractions generally cause MP to: A. Decrease B. Become zero C. Become unrelated D. Increase E. The answer is not in the choices	D — Stronger attractions generally require more energy to overcome.
5. Which size statement agrees with the PPT? A. Smaller always higher MP B. Size has no effect C. Size matters only for ions D. Larger always lower MP E. The answer is not in the choices	E — Larger molecules tend to have higher MP when forces are comparable.
6. Why use finely powdered sample? A. Raise molecular weight B. Load into capillary and observe melting C. Make it ionic D. Remove all IMFs E. The answer is not in the choices	B — The procedure calls for finely powdered sample in a capillary.
7. Why seal one end of the capillary? A. Hold sample at closed end B. Raise theoretical MP C. Measure molecular weight	A — The powder is compacted against the closed end.

D. Convert sample to vapor E. The answer is not in the choices	
8. The capillary is fastened to a: A. Balance B. Barometer C. Thermometer D. Volumetric flask E. The answer is not in the choices	C — The thermometer records the sample temperature.
9. What heats the sample in the procedure? A. Ice bath B. Paper towel C. Refrigerated water D. Well-stirred oil bath E. The answer is not in the choices	D — The PPT specifies a well-stirred oil bath.
10. T_1 is the temperature when the sample: A. Is completely liquid B. Begins to liquefy C. Is at room temperature D. Oil solidifies E. The answer is not in the choices	B — T_1 is the onset of melting.
PART II — MELTING-RANGE CALCULATIONS	
11. T_2 is the temperature when the sample: A. Begins melting B. Has not been heated C. Is completely liquefied D. Boils E. The answer is not in the choices	C — T_2 marks complete liquefaction.
12. Melting-range width is calculated by: A. $\Delta T = T_2 - T_1$ B. $\Delta T = T_1 - T_2$ C. $\Delta T = T_1 + T_2$ D. $\Delta T = T_1/T_2$ E. The answer is not in the choices	A — Subtract the onset temperature from the final melting temperature.
13. T_1=122°C and T_2=124°C. ΔT=? A. 246°C B. 2°C C. −2°C D. 1°C E. The answer is not in the choices	B — 124 − 122 = 2°C.
14. Salicylic acid: T_1=157°C, T_2=159°C. ΔT=? A. 316°C B. 157°C C. 1°C D. 2°C E. The answer is not in the choices	D — 159 − 157 = 2°C.
15. Oxalic acid: T_1=100°C, T_2=102°C. Melting range? A. 100–102°C B. 2–100°C C. 102–100°C D. 101–102°C E. The answer is not in the choices	A — The range runs from T_1 to T_2.
16. T_1=80°C and T_2=86°C. ΔT=? A. 166°C B. 3°C C. 6°C D. −6°C E. The answer is not in the choices	C — 86 − 80 = 6°C.
17. T_1=135°C and ΔT=3°C. T_2=? A. 132°C B. 138°C C. 405°C D. 135°C E. The answer is not in the choices	B — $T_2 = T_1 + \Delta T$ = 138°C.
18. T_1=T_2=149°C. ΔT=? A. 298°C B. 149°C C. −149°C D. 0°C E. The answer is not in the choices	D — 149 − 149 = 0°C.
19. Which has wider range: X=100–102°C or Y=121–126°C? A. X	E — X width is 2°C; Y width is 5°C; that comparison is not an option.

B. Equal C. Cannot calculate D. Neither E. The answer is not in the choices	
20. Benzoic acid values in the PPT: A. T_1=124, T_2=122, ΔT=2°C B. T_1=123, T_2=123, ΔT=2°C C. T_1=122, T_2=124, ΔT=2°C D. T_1=100, T_2=102, ΔT=2°C E. The answer is not in the choices	C — Benzoic acid T_1=122°C and T_2=124°C.
PART III — FORCES AND MOLECULAR INTERACTIONS	
21. What holds ions together in an ionic compound? A. Ionic bond B. Dispersion only C. H-bond only D. Dipole–dipole only E. The answer is not in the choices	A — Ionic compounds are held together by electrostatic attractions between ions.
22. Why do ionic compounds generally have high MP? A. No attractions B. Always small molecules C. All nonpolar D. Ionic bonds stronger than typical molecular IMFs E. The answer is not in the choices	D — Strong electrostatic attractions require substantial energy to overcome.
23. Common IMF order, strongest to weakest, in the PPT: A. Dispersion > H-bond > dipole B. H-bond > dipole–dipole > dispersion C. Dipole > dispersion > H-bond D. H-bond > dispersion > dipole E. The answer is not in the choices	B — This is the ranking presented in the PPT.
24. Which IMF occurs between polar molecules with dipole moments? A. Ionic only B. Metallic C. Dipole–dipole D. Nuclear E. The answer is not in the choices	C — Opposite partial charges on polar molecules attract.
25. As polarity increases, dipole–dipole interactions generally: A. Disappear B. Become ionic bonds C. Weaken D. Stop between molecules E. The answer is not in the choices	E — The PPT says greater polarity strengthens dipole–dipole interactions.
26. Hydrogen bonding is associated with H covalently bonded to: A. Small, highly electronegative F, O, or N B. Carbon only C. Any nonpolar molecule D. Like-charged ions E. The answer is not in the choices	A — The PPT identifies F, O, and N.
27. Why is hydrogen bonding unusually strong? A. H has largest radius B. H has no partial charge C. Electronegativity repels electrons D. H becomes significantly positive and approaches another atom's lone pair closely E. The answer is not in the choices	D — Partial charge and the small size of hydrogen strengthen the attraction.
28. Which statement about dispersion forces is correct? A. Only ionic compounds have them B. All molecules have them C. Only polar molecules have them D. Stronger than every H-bond E. The answer is not in the choices	B — Dispersion forces occur in all molecules.
29. More electrons generally cause dispersion forces to: A. Disappear B. Become ionic bonds C. Strengthen D. Become H-bonds E. The answer is not in the choices	C — The PPT states that more electrons mean stronger dispersion forces.
30. Intramolecular vs intermolecular forces: A. Intramolecular only in gases B. Intermolecular always stronger than covalent C. Same thing	E — Intramolecular forces act within a compound; intermolecular forces act between particles.

D. Intramolecular acts only between separate molecules E. The answer is not in the choices	
PART IV — SIZE, CHAIN LENGTH, AND BRANCHING	
31. How does the PPT estimate molecular size? A. Boiling time B. Molecular weight C. Color D. Number of samples E. The answer is not in the choices	B — Molecular weight is used as a practical estimate of size.
32. For comparable IMFs, a larger molecule generally has: A. Higher MP B. Lower MP always C. No surface area D. Ionic character necessarily E. The answer is not in the choices	A — Larger molecules tend to have more surface area in contact.
33. As carbon-chain length increases in a homologous series: A. Compounds become ionic B. Molecular weight decreases C. MP/BP always decrease D. MP and BP tend to increase E. The answer is not in the choices	D — The PPT gives this general trend for homologous series.
34. What is the stated effect of branching on alkane boiling point? A. Surface area increases without limit B. Alkane becomes ionic C. Shape becomes more spherical and surface area decreases D. BP always increases E. The answer is not in the choices	C — The PPT explains the boiling-point trend this way.
35. Safest statement about melting points of structural isomers? A. All isomers have identical MP B. More branching guarantees lower MP C. Molecular weight alone determines MP D. Structure is irrelevant E. The answer is not in the choices	E — Isomers can have different MPs; the PPT's branching explanation specifically addresses BP.
36. Which listed unbranched alkane has highest BP? A. Methane B. Decane C. Ethane D. Propane E. The answer is not in the choices	B — Decane has the highest listed BP, 174°C.
37. Which has highest listed MP among these unbranched alkanes? A. Methane, −182°C B. Ethane, −183°C C. Propane, −190°C D. Decane, −30°C E. The answer is not in the choices	D — −30°C is the highest/warmer MP among these choices.
38. Hexane's listed BP and MP in the C_6H_{14} table are: A. 60.3°C, −154°C B. 63.3°C, −118°C C. 68.7°C, −95°C D. 49.7°C, −98°C E. The answer is not in the choices	C — The table lists hexane as 68.7°C BP and −95°C MP.
39. Which C_6H_{14} isomer has the lowest listed BP? A. Hexane B. 2-Methylpentane C. 3-Methylpentane D. 2,3-Dimethylbutane E. The answer is not in the choices	E — 2,2-Dimethylbutane has the lowest listed BP, 49.7°C.
40. Which C_6H_{14} isomer has the lowest listed MP? A. 2-Methylpentane B. Hexane C. 2,2-Dimethylbutane D. 2,3-Dimethylbutane E. The answer is not in the choices	A — 2-Methylpentane has MP −154°C, the lowest listed.
PART V — COMPOUNDS AND EXPERIMENTAL RESULTS	
41. What is the listed melting point of NaCl? A. 158°C B. 801°C C. 123°C D. 101°C E. The answer is not in the choices	B — The PPT lists NaCl at 801°C.

42. Which order is highest to lowest MP for the four PPT compounds? A. Oxalic > benzoic > salicylic > NaCl B. Benzoic > NaCl > oxalic > salicylic C. Salicylic > oxalic > NaCl > benzoic D. NaCl > salicylic > benzoic > oxalic E. The answer is not in the choices	D — NaCl 801°C; salicylic acid ~158°C; benzoic acid ~123°C; oxalic acid ~101°C.
43. Salicylic acid's molecular formula is: A. $C_7H_6O_3$ B. $C_7H_6O_2$ C. $C_2H_2O_4$ D. NaCl E. The answer is not in the choices	A — The PPT lists salicylic acid as $C_7H_6O_3$.
44. Benzoic acid's approximate MP is: A. 801°C B. 158°C C. 123°C D. 101°C E. The answer is not in the choices	C — The PPT lists approximately 123°C.
45. Which statement about oxalic acid matches the PPT? A. Formula $C_7H_6O_3$ B. MP about 801°C C. Molar mass 138.121 g/mol D. Also called ethanedioic acid; formula $C_2H_2O_4$ E. The answer is not in the choices	D — Oxalic acid is ethanedioic acid, $C_2H_2O_4$, molar mass 90.03 g/mol.
46. Which groups are associated with the two H-bonds noted for salicylic acid? A. Two methyl groups B. –OH and –COOH C. Two chloride ions D. Two alkane chains E. The answer is not in the choices	B — The PPT notes two H-bonds associated with –OH and –COOH.
47. Which substance melted last in the PPT's experimental comparison? A. Salicylic acid B. Benzoic acid C. Oxalic acid D. Ethanol E. The answer is not in the choices	E — The PPT reports that NaCl melted last.
48. What does the PPT say about carboxylic acids? A. Strong H-bonding contributes to high MP and BP B. They contain no oxygen C. They cannot H-bond D. They are all ionic E. The answer is not in the choices	A — The PPT links strong hydrogen bonding with high melting and boiling points.
49. Which statement correctly distinguishes MP and BP? A. Same temperature for every compound B. MP is gas becoming liquid C. BP is liquid-to-gas; MP is solid-to-liquid D. BP is always lower than MP E. The answer is not in the choices	C — Melting is solid → liquid; boiling is liquid → gas.
50. Which combination summarizes the three main factors affecting MP in the PPT? A. Color, odor, solubility B. Label, lighting, container shape C. Density alone, volume, boiling time D. Bond holding particles, IMFs, molecular size E. The answer is not in the choices	D — The PPT lists bond type/particle attraction, intermolecular forces, and molecular size.